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Biomimetic Gap Analysis: Immune System Structural Patterns Applied to Agentic AI Security

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Abstract

As AI agents increasingly operate with autonomous access to enterprise identity infrastructure — holding credentials, calling APIs, making tool-use decisions — the security community lacks a coherent framework for detecting when these agents

are compromised or misbehaving. This paper introduces *biomimetic gap analysis*: a methodology that decomposes biological immune mechanisms across multiple kingdoms of life (mammals, plants, bacteria, insects) into abstract structural patterns and maps them against the structural patterns of current agent security architecture. The resulting gaps — where nature has a solution pattern but digital infrastructure does not — identify both novel detection approaches and attacker strategies that current defenses are blind to.

We present 34 cross-domain mappings from immune system mechanisms to agent security equivalents, 33 design principles for agent behavioral monitoring, and 16 biologically-derived attack scenarios. Key novel contributions include: the three-state detection model (calibrated-active, calibrated-missing, uncalibrated-inert) derived from corrected NK cell biology; the cell-autonomous security architecture derived from plant immunity; heritable threat memory derived from bacterial CRISPR systems; and the defense neutralization attack pattern derived from insect xenobiotic detoxification of plant toxins. We identify fundamental architectural gaps in current agent security — including the absence of infrastructure-sampled behavioral evidence, the absence of inherited security posture across agent generations, and the absence of self-defense capability in individual agents — and propose biologically-grounded approaches to address them.

Keywords: agentic AI security, artificial immune systems, biomimetic gap analysis, behavioral detection, identity infrastructure, autonomous agents, cross-domain structural patterns, multi-kingdom immunity

1. Thesis

Nature — specifically biological immune systems across multiple kingdoms of life — represents the largest dataset of solved adversarial optimization problems available. By decomposing immune mechanisms into abstract structural patterns and mapping them against the structural patterns of current agent security architecture, we can identify *gaps*: places where nature has a solution pattern but enterprise agentic infrastructure does not. Those gaps are candidate innovation spaces — both for novel detection approaches and for anticipating attacker strategies that current defenses are blind to.

This is not biomimicry in the surface-level sense ("the immune system has T cells, so build a T-cell-like thing"). It is *structural pattern transfer* [63, 64]: identifying the abstract relationship (e.g., "the default state is suspicion, not trust" or "detection and evasion co-evolve") and asking whether that pattern has a corresponding implementation — or a corresponding gap — in digital agent security. Existing biomimicry databases such as AskNature [60, 61] and FindStructure [62] organize biological strategies by function, but none systematically perform gap analysis against a specific

technology domain. A recent large-scale study found that fewer than 1,604 species have been explored in the entire biomimetics literature [65], indicating massive untapped potential.

1.1 Related Work and Positioning

Artificial Immune Systems (AIS). The application of immune system concepts to computer security is a mature field with a 40-year history. Farmer, Packard, and Perelson (1986) first suggested that computer science might borrow from the immune system [301]. Forrest, Perelson, Allen, and Cherukuri (1994) formalized this with their landmark paper "Self-Nonself Discrimination in a Computer," introducing the Negative Selection Algorithm [302]. Subsequent work produced the Clonal Selection Algorithm (de Castro & Von Zuben, 2002) [303], the Dendritic Cell Algorithm (Greensmith, Aickelin & Cayzer, 2005) [304], and the comprehensive textbook *Artificial Immune Systems: A New Computational Intelligence Approach* (de Castro & Timmis, 2002) [305]. AIS remains active, with recent work applying immune-inspired algorithms to IoT intrusion detection [306], industrial control systems [307], and semi-supervised learning for network security [308].

Schrom et al. (2023). The most directly relevant prior work is "Challenges in cybersecurity: Lessons from biological defense systems" (Schrom, Kinzig, Forrest, Graham, Levin, Bergstrom, et al., *Mathematical Biosciences*, 2023) [309]. This paper, authored by leading researchers in computational immunology (Forrest, Perelson), complex systems (Levin, Bergstrom), and cybersecurity (Halderman, Doupé, Rexford), presents a conceptual framework for structured comparisons between biological immunity and cybersecurity. It organizes defense mechanisms into five strategic layers, discusses the context of defense, and poses open research questions. Critically, Schrom et al. note that cloud-based computing containers — which "can interact, reproduce, and be destroyed, just like biological immune cells" — suggest that immune system principles should be revisited in light of modern computing architectures. They also observe that simpler immune systems (corals, bacteria) achieve multiple defensive layers, suggesting value in looking beyond vertebrate immunity.

Emerging agentic AI security frameworks. The application domain this paper targets — securing autonomous AI agents — is the subject of a growing body of work that approaches the problem without biological analogy. Neelou et al. (2025) introduce A2AS, a runtime security layer for AI agents that enforces certified behavior, authenticated prompts, security boundaries, in-context defenses, and codified policies [310]. Evani (2025) presents a control framework addressing adversarial propagation risks, multi-agent collusion, decision cascade exploits, and emergent behavior suppression in agentic AI systems [311]. Tutuncuoglu (2025) describes SentinelMind, a self-learning autonomous cybersecurity framework that draws loosely on biological immune systems and agent-based cognition for threat anticipation [312]. These

frameworks address the same problem domain as this paper but derive their architectures from engineering first principles rather than cross-domain biological analogy. Our work is complementary: the gap analysis and design principles presented here could inform the architectural evolution of frameworks like A2AS and Evani by identifying blind spots that engineering-first approaches may not surface.

Adversarial co-evolution and multi-agent threats. Tallam (2025) argues in "The Cyber Immune System" that adversarial forces — whether biological parasites or digital exploiters — should be reframed as essential stress-testers of complex systems, drawing on epidemiology and cybersecurity to examine how systems evolve through persistent threats [313]. This philosophical framing aligns with our Section 8 (evasion strategies), where pathogen evasion mechanisms inform attacker playbooks, though Tallam does not produce specific mappings or gap analysis. Separately, a growing body of work on multi-agent security independently identifies threat patterns that our biological analogies also surface: coordinated attacks that appear innocuous when viewed individually, steganographic communication between agents, and cascade-based threats that propagate across cooperative agent populations [314]. These overlap directly with our biofilm (#33), complement cascade (#8), and interferon (#23) mappings — providing independent engineering-side validation that the biological patterns transfer to real threat models.

How this paper differs from prior work. Our work builds on the AIS tradition and responds directly to the research agenda Schrom et al. propose, but differs in five specific ways:

(1) Multi-kingdom scope. AIS draws almost exclusively on mammalian adaptive immunity — primarily negative selection, clonal selection, and danger theory. Schrom et al. mention corals and bacteria in passing but do not systematically map their mechanisms. We draw on six kingdoms/domains (mammals, plants, bacteria, insects, pathogenic protists, viruses), producing mappings from plant cell-autonomous security, bacterial CRISPR heritable memory, insect transgenerational immune priming, insect melanization/encapsulation, caterpillar xenobiotic detoxification, trypanosome antigenic variation, and RNA interference — none of which appear in existing AIS literature applied to security.

(2) Systematic gap analysis. AIS papers typically propose immune-inspired algorithms. Schrom et al. propose a comparison framework and pose questions. We produce specific answers: 34 mechanism-to-security-equivalent mappings, each with an explicit GAP assessment identifying where the biological solution pattern has no digital counterpart. The gap column — not the mapping itself — is the primary novel contribution.

(3) Attack scenarios from pathogen evasion. AIS focuses almost entirely on defense. We systematically derive 16 attack scenarios from pathogen evasion strategies

— viral latency as sleeper agents, antigenic variation as identity rotation, biofilm formation as coordinated collective evasion, immune checkpoint exploitation as alert fatigue weaponization. The adversarial perspective is largely absent from prior AIS work.

(4) Applied to agentic AI security. AIS targets network intrusion detection (packets, system calls, network flows). We target autonomous AI agents operating with real credentials in enterprise identity infrastructure — a fundamentally different and newer problem domain that did not exist when AIS was founded.

(5) Structural pattern transfer, not algorithmic implementation. AIS builds executable algorithms. We extract abstract structural patterns and derive design principles. Our methodology is hypothesis-generating (identifying what should be investigated) rather than implementation-prescriptive (specifying what to build).

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2. Methodology

Mapping numbering convention. Mapping numbers (e.g., #1, #23, #32) are stable identifiers assigned during development and used as cross-references throughout the document. They do not reflect presentation order within sections. This preserves referential integrity across design principles, attack scenarios, and the coverage matrix.

1. Identify immune defense mechanisms across multiple species (mammals, plants, bacteria, insects).
2. For each mechanism, extract the **abstract structural pattern** — the species-independent logic of how the mechanism works.
3. Map each pattern to an **agent security equivalent** — a concrete implementation or design principle for monitoring and securing AI agents in enterprise identity infrastructure.
4. Identify the **gap** — does the equivalent exist in current practice? If not, that gap is either a novel detection opportunity or a blind spot an attacker could exploit.
5. Derive **attack scenarios** from pathogen evasion strategies — biological attackers that have evolved to defeat the defense mechanism suggest digital attack patterns.
6. **Vet** all biological claims against primary and authoritative secondary literature.

2.0.1 Why Biological Immunity Is the Right Source Domain

A cross-domain analogy requires justification: why should biological immune systems — rather than, say, military defense doctrine or financial fraud detection — be the source domain for agentic AI security?

The answer lies in a structural property shared by biological systems and non-deterministic AI that is absent from the domains where current security tools were designed. This connection was identified by Shaun Milligan via horticulture: determinate plant species (e.g., determinate tomatoes) grow to a fixed size, produce fruit, and stop — they have bounded, predictable behavior with a defined endpoint. Indeterminate species keep growing, keep producing, and have no built-in stop condition — their behavior is open-ended and variable. The parallel to AI is direct: deterministic AI systems produce the same output for the same input every time (bounded, predictable, like determinate plants), while non-deterministic AI systems (LLMs, stochastic agents) produce variable outputs for the same input (open-ended, like indeterminate plants).

This distinction has a critical security implication. Deterministic systems are easy to monitor: deviations from expected output are obvious signals. Traditional security tools — signature matching, rule-based IDS, allowlist/denylist enforcement — were designed for deterministic systems where "normal" has a fixed definition. Non-deterministic systems break these tools because you cannot draw a fixed boundary around "normal" when the output space is inherently variable. An agent that produces a different API call sequence each time it runs the same task is not malfunctioning — it is operating as designed. The monitoring challenge is not "did the output match the expected pattern?" but "is this variable output within the structural envelope of legitimate behavior?"

Biological immune systems evolved to solve exactly this problem. Biological organisms are indeterminate systems — cells divide, differentiate, mutate, and adapt continuously. The immune system does not monitor for deterministic signatures of health; it monitors for structural invariants and behavioral patterns that hold even when the specific behavior varies. MHC-I does not check whether a cell produced the "correct" protein — it samples whatever the cell IS producing and asks whether it falls within the organism's self-repertoire. This is monitoring by structural envelope, not by deterministic output matching.

This is why biological immunity — rather than military doctrine or financial fraud detection, which evolved for more deterministic threat models — is the right source domain for non-deterministic agent security. The immune system has solved the monitoring problem for indeterminate systems. Current security tooling has not.

2.1 Methodological Constraints and Disanalogies

The locality constraint problem. Most immune system strategies depend on a physical property that digital systems do not share: locality. Biological threats must physically travel between cells. Signaling molecules diffuse at chemical speeds. Immune cells must physically migrate to infection sites. Digital systems have no equiv-

alent constraint — a compromised agent can exfiltrate data at network speed to any endpoint on the internet.

Implications:

- **Mappings based on information asymmetry patterns transfer well.** Detection approaches — behavioral sampling, missing-self recognition, complementary detector coverage, threat memory — do not depend on locality.
- **Mappings based on containment patterns transfer with caveats.** Inflammation (dynamic scope narrowing) works biologically because the affected region is physically bounded. Digitally, preemptive scope restriction may be more appropriate than reactive containment.
- **Any mapping that assumes the attacker is constrained to local or sequential action needs redesign.** Digital attackers can act on all resources simultaneously.

The analogical reasoning caveat. Structural similarity between domains is a heuristic for generating hypotheses, not a proof that solutions transfer. Every mapping in this document should be read as "the immune system's approach suggests we should investigate whether…" not "agents should be required to…"

2.1.1 Falsifiability Criteria

A mapping in this framework can be DISPROVED — and should be discarded or revised — under any of the following conditions:

The gap claim is wrong. If the capability described as "absent" in the gap column demonstrably exists in a shipping agent security framework and performs the function described, the gap is closed and the mapping's novel contribution is reduced to documentation of existing practice. Gap claims in this paper are practitioner assessments, not exhaustive surveys; they are the most likely failure point.

The structural pattern does not survive implementation. If a prototype implementation of the agent security equivalent reveals that the biological pattern depends on a physical property (locality, chemical diffusion, cellular compartmentalization) that has no digital analog AND no digital workaround, the mapping fails at the transfer step. The locality constraint (Section 2.1) is the most common cause.

The attack scenario is mechanically infeasible. If an attack scenario derived from a pathogen evasion strategy cannot be executed against any real agent framework — not because defenses prevent it, but because the attack mechanism has no digital equivalent — the scenario should be downgraded from "anticipated threat" to "theoretical only."

The design principle produces worse outcomes than no principle. If implementing a design principle causes more damage than the threat it addresses (false

positive cascades, unacceptable performance overhead, operational complexity that exceeds defensive value), the principle fails the cost-benefit test that biological immune systems are subject to through natural selection.

2.2 Key Definitions

Agent (in this document): An AI system operating with autonomous access to enterprise infrastructure — holding credentials (OAuth tokens, API keys, service account identities), calling APIs, making tool-use decisions, and interacting with other systems on behalf of a human principal or an automated workflow. Distinct from "agent" in the biological sense (pathogenic agent) or security tooling sense (monitoring agent), though both usages appear in source literature.

Behavioral signature: A standardized, infrastructure-captured, structured record of a single unit of agent action, collected at the tool/resource/credential interaction layer (not self-reported by the agent). Format: a tuple of (agent_id, tool_called, target_resource, credential_used, timestamp, payload_hash, session_context). Analogous to the peptide fragment displayed on MHC-I — a short, structured representation of one unit of internal activity. Analyzable as a distribution over time. Comparable across agents via standardized format.

Structural pattern: The species-independent, substrate-independent logic of how a biological mechanism works. Not the specific molecular implementation, but the abstract relationship it embodies. Example: "silence is the alert signal" (the structural pattern underlying NK cell missing-self detection) can be implemented in biology (MHC-I absence → NK activation) or in digital infrastructure (telemetry absence → elevated monitoring) using completely different substrates.

Gap: A place where the biological solution space has a structural pattern that the digital agent security space does not. A gap is a hypothesis — it suggests either (a) a novel detection/defense approach that should be investigated, or (b) a blind spot that an attacker could exploit, or (c) both simultaneously.

Biomimetic gap analysis: The methodology introduced in this paper. Decompose immune mechanisms into abstract structural patterns. Map those patterns against the structural patterns of current agent security architecture. Identify gaps. Derive both novel defenses and novel attack scenarios from the gaps.

Pathobiont: An indigenous microorganism (typically a gut commensal bacterium) that is normally benign or beneficial but can promote disease under certain changed conditions — such as disruption of the host immune system, alteration of the microbial community, or breach of the epithelial barrier. Distinct from a pathogen (which is inherently harmful) and from a commensal (which remains benign). The patho-

biont concept is central to Mapping #15 and Attack Scenario #7, where a previously tolerated third-party integration becomes a threat under changed conditions.

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3. Mappings: Detection and Recognition

Mapping #1–2 (Consolidated): MHC-I Self-Declaration + NK Cell Missing-Self Detection

Source organism: Mammals

Biology: All nucleated cells continuously display peptide fragments of their internal proteins on MHC class I molecules on their surface [1, 2]. Cytotoxic T cells scan these displays for abnormal peptides. Separately, NK cells monitor for the PRESENCE of MHC-I — cells that stop displaying MHC-I (which viruses cause in order to hide from T cells) are killed by NK cells [3].

Corrected model (three states, not two):

- **Calibrated + Inhibited = Tolerant.** NK cell has inhibitory receptors matching self-MHC, is receiving inhibitory signals from healthy cell. Cell is actively monitored and confirmed healthy [4, 5].
- **Calibrated + NOT Inhibited = Kill.** NK cell is educated but encounters a cell missing MHC-I. Absence of expected signal triggers attack [4].
- **Uncalibrated = Inert/Hyporesponsive.** NK cells lacking inhibitory receptors for self-MHC do not become indiscriminate killers — they become functionally inert [4, 5, 6]. An uneducated detector does nothing, not everything [7].

Abstract pattern: Every legitimate entity must continuously broadcast proof of its internal operations. Two independent detection systems with complementary blind spots monitor these broadcasts — evading one system triggers the other. Uncalibrated detectors are useless (inert), not dangerous (over-aggressive). The system fails safe toward INACTION when uncalibrated, and toward ACTION when calibrated but not receiving expected signals.

Agent security equivalent: Agents emit structured telemetry from the infrastructure layer (not self-reported logs). A monitoring layer watches for abnormal telemetry. A SEPARATE monitoring layer watches for ABSENT telemetry — agents that stop emitting are flagged by the second system even if the first sees nothing wrong.

Gap: Identity systems do not monitor for the ABSENCE of expected identity assertions. If an agent stops presenting tokens, most systems see nothing. No session, no alert. The agent disappears from the identity plane. Additionally, behavioral monitoring tools deployed without adequate baselining are inert, not aggressive — they give false confidence that monitoring is occurring.

Mapping #3: Antigen Presentation (Infrastructure-Sampled Evidence)

Source organism: Mammals

Biology: MHC class I molecules display peptide fragments generated by the proteasome — a systematic quality-control process that degrades ubiquitin-tagged proteins [2, 8]. The cell does not choose which peptides to display. The monitoring system captures waste products of normal operations, not curated reports.

Abstract pattern: The monitored entity does not choose what evidence is inspected. The monitoring infrastructure independently samples the entity's actual operations — intercepting tool calls, inspecting payloads, sampling memory state — without the entity's cooperation or awareness.

Agent security equivalent: Behavioral signatures should be captured at the infrastructure layer (API gateway, token exchange service, tool execution runtime), not from the agent's own log output. A compromised agent can falsify its own logs. It cannot falsify what the API gateway observed.

Gap: Current observability for agents is almost entirely self-reported. There is no independent "proteasome" that samples from the agent's actual execution environment and surfaces results to an external inspector.

Mapping #4: Pattern Recognition Receptors / PAMPs (Innate vs. Adaptive — Two-Speed Detection)

Source organism: Mammals

Biology: Innate immunity uses germline-encoded pattern recognition receptors (TLRs, etc.) to detect conserved molecular patterns shared by groups of pathogens (PAMPs). This is fast, generic, and has no memory. Adaptive immunity (T/B cells) is slower, specific, and remembers past threats [1, 9, 70].

Abstract pattern: Fast/generic detection runs in parallel with slow/specific detection. The fast layer buys time for the slow layer to build a precise response. The slow layer remembers past threats for faster future response.

Agent security equivalent: A fast heuristic layer (rate limiting, known-bad pattern matching, basic behavioral bounds) runs alongside a slower analytical layer that builds agent-specific behavioral profiles over time and flags deviations from learned baselines.

Gap: We have rate limiting and WAFs (innate-ish). We lack adaptive behavioral profiling for individual agents. UEBA exists for humans but has not been adapted for agentic workloads.

Mapping #12: Dendritic Cells (Pervasive Triage-and-Transport)

Source organism: Mammals

Biology: Dendritic cells patrol boundary surfaces, collect samples of suspicious material, and physically carry those samples to lymph nodes where the adaptive immune response is coordinated [1, 24]. They decide what to escalate and where to bring it, not what to do about it.

Abstract pattern: Lightweight monitoring agents at the boundary of each agent environment collect behavioral samples and route them to a centralized analysis engine with deeper analytical capability. Boundary monitors are pervasive and good at sampling. Central analysis is smart but cannot be everywhere.

Agent security equivalent: API gateways, tool-call interceptors, and token exchange points should collect behavioral samples and route suspicious patterns to an analysis tier — not just check authentication and rate limits.

Gap: API gateways exist but do not perform behavioral sampling. They do not take request payload samples, compare to behavioral models, and route anomalies to analysis.

Mapping #23: Interferon Response (Preemptive Neighbor Warning)

Source organism: Mammals

Biology: When a cell detects viral infection, it secretes type I interferons (IFN-α/β) that bind to receptors on neighboring uninfected cells, activating the JAK-STAT pathway and inducing hundreds of interferon-stimulated genes (ISGs) that establish a preemptive antiviral state [181, 183, 188]. The signal is amplified through a positive feedback loop — each cell that receives interferon also produces more interferon, spreading the defensive activation ahead of the threat [183]. Nearly every cell type can both produce and respond to interferons [185]. The result: by the time the pathogen reaches the next cell, that cell has already activated hundreds of defensive genes [182, 188].

Abstract pattern: When an entity detects a threat, it automatically broadcasts a defensive signal to all neighboring entities. Recipients preemptively activate their own defenses BEFORE they are attacked. The signal self-amplifies — each recipient

also broadcasts, creating a wave of preemptive hardening that moves faster than the threat itself. This is peer-to-peer and automatic, not centrally coordinated.

Agent security equivalent: When an agent detects a threat signal (anomalous input, unexpected credential in context, impossible tool call results), it automatically broadcasts a structured defensive signal to every agent it communicates with. Recipients preemptively tighten their defenses: reduce permission scope, increase telemetry emission, activate additional input validation, re-authenticate credentials. Each recipient re-broadcasts. The hardening wave propagates at network speed.

Gap: No agent framework emits defensive signals to other agents. Agents operate in isolation. If Agent A detects something suspicious, Agent B has no way of knowing. The closest analog is a centralized SIEM — but that requires both agents to report, the SIEM to correlate, and the SIEM to push directives back. The interferon model is peer-to-peer and automatic.

Distinction from Mapping #17 (Plant SAR): Both describe preemptive neighbor warning, but differ in scope and mechanism. Interferon (#23) is FAST, protein-mediated, cell-to-cell within a single organism, and triggers hundreds of specific defensive genes in recipients. Plant SAR (#17) is SLOWER, chemical-mediated (salicylic acid), travels systemically through the vasculature, and even emits volatile signals that warn NEIGHBORING PLANTS — extending the warning across organizational boundaries. The security design implication: interferon maps to intra-fleet agent-to-agent signaling; SAR maps to cross-organization threat intelligence sharing.

Mapping #32: RNA Interference (Detection IS the Weapon)

Source organism: Plants, invertebrates (*Drosophila*, *C. elegans*), fungi

Biology: When a cell detects double-stranded RNA — a signature of viral replication that does not normally occur in eukaryotic cells — the Dicer enzyme chops it into small interfering RNAs (siRNAs) of 21–25 nucleotides [271, 274]. These siRNAs are loaded into the RISC complex, where they serve as guide strands: each siRNA base-pairs with complementary viral mRNA, directing its destruction [271, 272]. The detection event itself produces the guided weapon. In plants, siRNAs can spread from cell to cell, conferring resistance on neighboring cells before the virus reaches them [273]. Viruses have counter-evolved suppressors of RNA silencing (VSRs) that block Dicer, sequester siRNAs, or inhibit RISC assembly [276].

Abstract pattern: The detection system does not merely DETECT the threat and then invoke a separate response mechanism. Instead, detection and destruction are the SAME process: the act of recognizing the threat's signature automatically produces the precision-guided weapon that destroys the threat. The signature fragment

IS the targeting mechanism. And the weapon can propagate to neighboring entities.

Agent security equivalent: When a behavioral anomaly is detected in an agent's API call pattern, the detection system should automatically generate a BLOCKING RULE derived directly from the detected pattern — not just an alert for a human to write a rule later. The detection signature becomes the filter. If an agent makes an unauthorized API call to `exfil.evil.com/data`, the detection of that call should AUTOMATICALLY produce a network rule blocking `exfil.evil.com` and propagate it to all peer agents. Detection and response collapse into a single operation.

Gap: Detection and response are almost always separate workflows operated by separate teams. SIEM detects → ticket → SOC reviews → engineer writes rule → rule deployed. The RNAi model says: the act of detection should produce the response as a direct byproduct, not as a downstream workflow.

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4. Mappings: Response and Escalation

Mapping #8: Complement Cascade (Escalating Tag-and-Respond)

Source organism: Mammals

Biology: A small detection signal triggers a multiplicative cascade [14, 17]. The initial signal tags the suspect entity. Tags recruit more taggers. Accumulation of tags escalates the response from "investigate" to "contain" to "destroy." Legitimate host cells actively express regulatory proteins that REPEL tag accumulation [15, 16].

Abstract pattern: The system has four states — verified healthy (actively attesting), unchecked (no signal either way — treat as suspicious), tagged (under elevated monitoring with reduced privileges), and contained (isolated, pending investigation). Legitimate entities actively produce anti-tags (health attestations). An entity with no tags AND no health signals is in the most dangerous state: unchecked.

Agent security equivalent: Suspicious behavior should TAG an agent — increase monitoring fidelity, reduce privilege scope, notify adjacent monitoring systems. Accumulating tags trigger escalation to containment. Legitimate agents should continuously express health signals (regular attestation, challenge-response, output integrity verification).

Gap: Current security posture is mostly binary: allowed or blocked. There is no intermediate "tagged for elevated scrutiny, still running, privileges narrowing." No concept of agents actively expressing health attestations to prove they are clean.

Mapping #13: Inflammation (Dynamic Containment Through Accepted Collateral Damage)

Source organism: Mammals

Biology: When a threat is detected in a region, the body deliberately degrades local performance (increased vascular permeability, tissue swelling, elevated temperature) to contain the threat's spread [1, 15]. This disrupts normal function in the affected area. The cost of containment is accepted as lower than the cost of unchecked spread.

Abstract pattern: When suspicious agent behavior is detected in a particular environment, automatically degrade the performance envelope of that environment — throttle API rates, narrow network access, increase latency on cross-environment calls. Accept the cost to legitimate operations.

Agent security equivalent: Dynamic, reactive scope narrowing in response to behavioral anomaly signals. Boundaries tighten when threat signals are detected and relax when cleared.

Gap (with locality caveat): Network segmentation is static. Dynamic, reactive degradation in response to behavioral signals is mostly absent. However, due to the locality constraint problem (Section 2.1), digital "inflammation" must be PREEMPTIVE (narrow default scopes that agents earn their way out of) rather than purely reactive.

Mapping #18: Hypersensitive Response (Self-Sacrifice Containment)

Source organism: Plants

Biology: When a plant cell detects a pathogen, it kills ITSELF and the surrounding cells to create a dead zone the pathogen cannot cross [41, 43]. Local, immediate, destructive — no waiting for central coordination.

Abstract pattern: When an entity detects confirmed compromise, it immediately terminates itself AND degrades its immediate environment to prevent the threat from using the entity's resources to spread.

Agent security equivalent: Agents should have a hardwired capability to revoke their own credentials, terminate their own process, and invalidate sessions they have established with other systems. Agent-level "dead man's switch" security, pre-authorized to self-destruct.

Gap: No agent framework currently supports self-termination based on self-assessed compromise. Agents lack detection logic to know they are compromised, and lack the ability to revoke their own credentials on the way out.

Mapping #22: Apoptosis vs. Necroptosis (Quiet Death vs. Noisy Death)

Source organism: Mammals

Biology: The body has two modes of killing compromised cells. Apoptosis is "silent" — the cell packages itself neatly into apoptotic bodies that are cleaned up by phagocytes without triggering inflammation [171, 172]. Necroptosis is "noisy" — the cell ruptures, releasing alarmins and proinflammatory signals that alert the surrounding immune system [171, 176]. Necroptosis serves as a BACKUP: when viruses block apoptosis (to keep their host cell alive), the cell switches to necroptosis, guaranteeing the immune system is alerted even if the quiet path is blocked [174, 176]. These are not independent pathways — they are interconnected tools generating a spectrum of immunogenic outcomes [171].

Abstract pattern: The system has two termination modes — quiet (cleans up without systemwide alert) and noisy (releases alarm signals triggering broader response). Quiet is preferred when the threat is contained. Noisy is used when quiet termination is BLOCKED by the attacker. If the attacker prevents quiet termination, the system automatically escalates to noisy. The attacker cannot prevent BOTH.

Agent security equivalent: Quiet termination (apoptosis): graceful shutdown — revoke credentials, close sessions, drain operations, archive logs, notify only direct monitoring. Used for routine decommissioning or localized issues. Noisy termination (necroptosis): hard kill — immediately revoke all credentials, terminate all sessions, emit structured alarm, trigger incident response, flag all interacting systems. Used when compromise is confirmed or when the agent RESISTS quiet termination.

Gap: Agent termination today is almost always quiet with no automatic escalation to noisy. If credential revocation fails, it retries silently. If an agent process ignores SIGTERM, the system waits. There is no mechanism that says "if this agent didn't die when we told it to, THAT IS AN ALARM." The failure of quiet termination should trigger noisy termination.

Mapping #28: Clonal Expansion (Dynamic Scaling of Threat-Specific Detection)

Source organism: Mammals (B cells, T cells, NK cells)

Biology: Before antigen exposure, the number of lymphocytes specific for any particular antigen is between 1 in 10^5 to 10^6 [231]. When a specific detector matches an actual threat, it undergoes massive clonal expansion — rapidly proliferating into

thousands of identical copies [231, 233]. Some clones differentiate into effector cells (immediate threat response) while others become long-lived memory cells (preparation for the same threat's return) [233, 236]. Even innate lymphocytes like NK cells undergo clonal expansion during infection [234]. The amplification is proportional to the strength of the match — stronger binding produces larger expansions [233].

Abstract pattern: The system maintains a massive, diverse library of detection capabilities at low copy numbers. When one specific detector matches a real threat, that detector is MASSIVELY amplified — producing thousands of copies. Some copies become immediate effectors (fight the current threat). Others become long-lived memory (prepare for the same threat's return). Amplification is proportional to match severity.

Agent security equivalent: When a specific detection rule fires against a confirmed threat, the monitoring infrastructure dynamically scales up that rule's deployment across the entire agent fleet at high priority with dedicated compute. Amplification is proportional to confidence: a confirmed true positive triggers massive scale-up; a low-confidence alert triggers modest scale-up. Some scaled-up instances are "effector" (actively blocking/alerting in real-time) while others are "memory" (persisting long after the incident at low priority, ready to scale up instantly if the same threat returns).

Gap: Detection rules are deployed uniformly regardless of whether they have matched real threats. A rule that just caught a real attacker gets no more resources than a rule that has never matched anything. There is no mechanism to dynamically amplify a proven detection rule or to persist amplified rules as "memory" capacity after the incident resolves.

Mapping #34: Fever (Systemic Environmental Degradation)

Source organism: Mammals, birds, reptiles, fish, invertebrates

Biology: When infection is detected, the hypothalamus raises the body's core temperature by 1–4°C [291, 293]. This elevated temperature directly inhibits the growth of many pathogens (which are optimized for normal body temperature), enhances immune cell activity (neutrophil migration, T cell proliferation, cytokine production), and increases the rate of iron sequestration from the blood (depriving pathogens of a critical nutrient) [291, 294]. The host pays a significant metabolic cost — a 1°C rise in body temperature requires a 10–12.5% increase in metabolic rate [291, 293] — but the cost is accepted as lower than unchecked infection. The fever response has been conserved across vertebrates and invertebrates for over 550 million years of evolution [292, 294]. Unlike inflammation (which is local), fever is SYSTEMIC — it affects the entire organism.

Abstract pattern: When a threat is confirmed, the system deliberately degrades its OWN operating environment across the ENTIRE infrastructure — not just locally. The degradation makes conditions worse for the attacker while remaining barely tolerable for legitimate operations. The cost to the system is significant but accepted as lower than the cost of unchecked compromise.

Agent security equivalent: When a confirmed compromise is detected anywhere in the agent fleet, SYSTEMWIDE hardening activates: all agents have their permission scopes reduced, all API rate limits tightened, all credential TTLs shortened, all monitoring sensitivity increased. Every agent in the fleet operates at degraded performance — slower, more constrained, more monitored — until the threat is resolved. Legitimate operations suffer, but the attacker's ability to move laterally, escalate privileges, or exfiltrate data is dramatically reduced.

This is distinct from inflammation (Mapping #13), which is LOCAL — narrowing scope in the affected region. Fever is SYSTEMIC — degrading the entire environment. The inflammation model says "quarantine the affected zone." The fever model says "make the entire infrastructure hostile."

Weaponized-triggering caveat: Automated systemwide hardening is a double-edged sword. An attacker who can deliberately trigger the fever response — for example, by generating a convincing false-positive compromise signal — causes self-inflicted damage across the entire infrastructure. This maps directly to the cytokine storm failure mode (Section 13): an over-response that causes more damage than the original threat. A digital fever response MUST have safeguards: confirmation thresholds requiring multiple independent signals before systemwide activation, graduated severity levels, maximum duration limits, and manual override capability.

Gap: Systemwide hardening in response to confirmed compromise is almost never automated. Incident response teams may manually tighten controls during a major incident, but there's no automated "fever response" that immediately degrades the operating environment fleet-wide. The fever model says: this should be a pre-configured, automated response triggered by confirmed compromise detection — with safeguards against weaponized triggering.

Mapping #31: Melanization/Encapsulation (Hostile Containment Shell)

Source organism: Insects (*Drosophila*, *Manduca sexta*, *Anopheles*)

Biology: When an insect encounters a pathogen too large to phagocytose (such as a parasitoid wasp egg), hemocytes surround the invader in multiple layers, forming a capsule [261, 263]. The capsule is then melanized — the enzyme phenoloxidase generates toxic quinones, reactive oxygen species, and melanin that are concentrated

INSIDE the capsule [261, 263, 265]. The encapsulated pathogen is killed by the hostile chemical environment while the capsule is presumed to protect the host from those same toxic compounds by acting as a containment barrier [261]. The melanization reaction is strictly localized — it occurs only on the surface of the intruder, and the hemocyte layers separate the toxic microenvironment from host tissue [263, 265].

Abstract pattern: When a threat is too large or complex to eliminate directly, the system physically ENCAPSULATES it in a hostile microenvironment. The containment shell serves two simultaneous functions: (1) concentrating toxic countermeasures on the intruder, and (2) protecting the host from the collateral damage of those countermeasures. The containment is active — it generates increasingly hostile conditions inside while maintaining safety outside.

Agent security equivalent: When a compromised agent or suspicious process is identified but cannot be immediately terminated (dependencies, data recovery needs, forensic preservation), place it in a "melanization shell": a network segment with progressively tightening constraints. Inside the shell: all outbound connections are proxied through content inspection, all credentials are replaced with monitored honeytokens, all API responses are delayed and logged, all data writes are shadowed. The agent continues to operate — and if the containment is sufficiently transparent, the attacker may not immediately recognize they are contained — but the environment inside the shell becomes progressively more hostile: more monitored, more constrained, more deceptive. Meanwhile, the shell protects the rest of the infrastructure from whatever the agent does inside. (Caveat: a sophisticated attacker may probe for containment signals — latency changes, honeytoken responses, network topology differences — so melanization shells must be designed to resist detection, not assumed to be invisible.)

Gap: Containment today is binary — either the agent runs freely or it's terminated. There is no intermediate state of "actively hostile containment" where the agent continues operating but in a progressively degrading and deceptive environment. The melanization model says: containment should be an active process that generates intelligence, not just a wall.

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5. Mappings: Learning, Memory, and Calibration

Mapping #9: Thymic Education (Training Detectors Before Deployment)

Source organism: Mammals

Biology: T cells undergo positive and negative selection in the thymus [18, 19, 23]. ~98% die [18]. Positive selection ensures T cells can recognize MHC (functional calibration). Negative selection eliminates T cells that over-react to self-antigens (pre-

venting autoimmunity) [20, 21]. The thymic medulla expresses proteins from all over the body via the AIRE gene so T cells are trained on comprehensive "normal" BEFORE deployment [19, 22].

Abstract pattern: Before a detection entity is deployed, it goes through a training environment with comprehensive examples of normal behavior from across the entire system. Detectors that cannot recognize normal are eliminated (useless). Detectors that over-react to normal are eliminated (dangerous — autoimmune / alert fatigue). Only calibrated detectors survive. The training corpus must include representative behaviors from every agent type in every environment.

Agent security equivalent: Behavioral monitoring systems need a comprehensive baselining phase before deployment — covering legitimate agent behaviors across the entire enterprise, not just the local environment.

Gap: Fundamentally absent for agentic workloads. There is no "thymus for agent monitors." No standardized training phase, no comprehensive corpus of legitimate agent behaviors, no culling of over-sensitive or under-sensitive detection rules before deployment. Detection is deployed live and tuned reactively.

Evasion implication — Training Data Poisoning (Attack Scenario #4): If an attacker injects malicious behavior patterns into the "normal" baseline during the learning phase, the monitoring system will classify those patterns as self and never flag them.

Mapping #5: Immune Memory (Distributed Threat Signature Catalog)

Source organism: Mammals

Biology: After a threat is neutralized, memory T and B cells retain compressed representations of the threat's signature [1, 9]. These memory cells circulate throughout the body (distributed) and also reside in specific tissues (locally cached). Future encounters produce faster, stronger responses.

Abstract pattern: When a threat is detected and neutralized, the system generates a compressed behavioral signature and distributes it to all monitoring nodes. Memory is both globally distributed AND locally cached at high-risk boundary points.

Agent security equivalent: Threat behavioral signatures should propagate in real time from the detection point to all monitoring nodes across all environments. Not a centralized threat database alone — a distributed, agent-level memory.

Gap: We have centralized threat intel (STIX/TAXII, IOCs). We lack distributed, agent-specific behavioral signatures that propagate automatically when a pattern is detected in one environment.

Mapping #20: CRISPR-Cas Adaptive Immunity (Genetic Write-Once Threat Memory)

Source organism: Bacteria

Biology: When bacteria encounter a phage, they capture a short DNA fragment from the phage and integrate it into their CRISPR locus — a permanent, heritable memory bank [47, 49]. All descendant bacteria inherit all accumulated spacers [47, 48]. On re-encounter, the stored spacer is transcribed into a guide RNA that directs Cas nucleases to cleave matching foreign DNA [48, 52]. A PAM (protospacer-adjacent motif) mechanism prevents the system from attacking its own stored signatures (self/non-self discrimination) [47]. When a phage mutates to evade an existing spacer, the system ACCELERATES acquisition of new spacers (primed acquisition) [48, 50]. Note: the characterization of CRISPR as "Lamarckian evolution" [47] is debated — some aspects are better described as Darwinian selection on variants that acquired spacers.

Abstract pattern: When a threat is encountered, the system captures a precise signature and writes it permanently into a shared, heritable, append-only memory bank. All descendant entities inherit all signatures. Exact match triggers immediate destruction. Near-miss triggers ACCELERATED learning. A self/non-self marker prevents false positives against stored signatures.

Agent security equivalent: Every agent should be both a consumer and a producer of threat intelligence. When an attack pattern is detected, a structured signature is captured and written into the agent deployment infrastructure. All subsequently deployed agents inherit all signatures. Near-miss detection triggers automatic generation of variant signatures. Context markers (environment, confidence, false-positive rate) prevent legitimate traffic from being blocked by overly broad signatures.

Gap: Threat intelligence is consumed (subscribed to), not generated at the point of attack and automatically propagated to the agent fleet. Incident signatures are not automatically inherited by new deployments. Evasion does not trigger faster learning.

Mapping #21: Transgenerational Immune Priming (Inherited Security Posture)

Source organism: Insects (bumblebees, flour beetles, tobacco hornworm), plants

Biology: Parents who encounter pathogens can prime their offspring's immune systems through two channels [53, 54]: (1) Maternal — specific — physical transfer of pathogen fragments into eggs, pre-arming offspring against specific threats [56, 58].

(2) Paternal — general — epigenetic modifications (DNA methylation, histone acetylation) that elevate general immune readiness without reference to specific pathogens [53, 55]. This transgenerational priming has been demonstrated in bumblebees, flour beetles, tobacco hornworm, and other invertebrates [54, 57], as well as in plants [59].

Abstract pattern: When an entity encounters a threat, it produces two types of inheritable security information: (1) SPECIFIC threat signatures baked into the successor's initial configuration, and (2) GENERAL security posture elevation — configuration changes that increase overall defensive readiness (higher logging, tighter permissions, more aggressive anomaly detection) without targeting any specific threat. Both channels are automatic — triggered directly by the incident, not by human intervention.

Agent security equivalent: When a security incident occurs, the CI/CD pipeline for agent deployments should AUTOMATICALLY incorporate both specific signatures (blocklists) and general hardening (template modifications) into all subsequent deployments. The feedback loop from "threat encountered" to "next generation is harder to kill" should be hours, not months.

Gap: Incident response and deployment engineering are separate workflows, teams, and toolchains. Agent deployment configurations are almost never modified as a direct consequence of security incidents. The inheritance mechanism is manual and lossy.

Mapping #24: Affinity Maturation (Live-Tuning Detection During Active Incidents)

Source organism: Mammals

Biology: During an active immune response, B cells in germinal centers undergo rapid somatic hypermutation — random point mutations in their antibody genes — followed by competitive selection where only cells producing higher-affinity antibodies survive [191, 192, 198]. This iterative mutate-test-select cycle produces antibodies of successively greater affinity for the specific pathogen being fought. Critically, high-performing B cells are protected: they divide more but mutate LESS per division, preserving what's working [191]. Recent research shows this process can even generate entirely new detection capabilities that didn't exist before the encounter [196]. (Note: The regulated mutation rate finding [191] and the de novo specificity generation finding [196] are both from 2025 publications. The core mechanism of somatic hypermutation and affinity maturation is well-established; these specific refinements are recent and may be updated as the field develops.)

Abstract pattern: During an active threat encounter, the detection system runs a

rapid mutation-and-selection cycle: duplicate detectors, randomly vary their parameters, test variants against the actual threat, KEEP better performers, KILL worse performers. Each cycle produces more precise detection. The system gets BETTER at recognizing the specific threat WHILE fighting it. High-performing detectors are protected from excessive modification.

Agent security equivalent: During an active security incident, enter "germinal center" mode: take the behavioral signature that triggered the initial alert, generate VARIANTS by modifying detection parameters (time windows, threshold values, API call sequence patterns), test each variant against the observed data stream, score by precision, keep best performers, discard rest. Iterate. By incident end, the detection rule is dramatically more precise. Rules already performing well should be locked and protected from further modification.

Gap: Detection rules are not iteratively improved during active incidents. SOC analysts manually tune rules post-incident. There is no automated mutation-and-selection cycle that refines rules in real time based on data flowing through during an active incident.

Mapping #25: Trained Immunity (Dumb Detectors That Learn)

Source organism: Mammals

Biology: Innate immune cells — which were supposed to be generic, memoryless first responders — can be epigenetically reprogrammed by previous encounters to mount enhanced responses to future threats [201, 204]. This "trained immunity" is mediated through histone modifications (H3K4me1, H3K4me3, H3K27ac) at inflammatory gene promoters, making those genes easier to activate [201, 207]. The training is not specific to any one pathogen — it's a general elevation of readiness [204]. BCG vaccination, for example, trains monocytes and macrophages to respond more vigorously to completely unrelated pathogens [202, 207]. And the reprogramming extends to bone marrow precursor cells — all FUTURE innate immune cells inherit the enhanced responsiveness [201, 205].

Abstract pattern: "Dumb" first-line detection systems that were designed as generic and stateless can be reprogrammed by experience to respond more vigorously to future threats. The reprogramming is general (not threat-specific), persists long-term, and is inherited by new instances. The clean distinction between "dumb stateless detection" and "smart stateful detection" is false — even the dumb layer can learn.

Agent security equivalent: WAFs, API rate limiters, basic pattern-matching rules — the "innate" security layer — should be able to globally elevate their sensitivity after encountering threats. Not by adding specific rules (that's adaptive/SIEM ter-

ritory) but by AMPLIFYING their baseline responsiveness. A WAF that processed traffic during an active attack could increase scoring sensitivity for ALL subsequent traffic. A rate limiter triggered by credential stuffing could tighten thresholds for ALL endpoints. This training should propagate to new instances — when a new WAF is deployed, it inherits the trained state of existing WAFs, not factory defaults.

Gap: First-line security tools are completely stateless. Every WAF restart, every new instance deployment starts from factory configuration regardless of what previous instances experienced. The experience of the innate security layer is discarded on every restart. The trained immunity model says that's a massive missed opportunity.

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6. Mappings: Regulation and Suppression

Mapping #11a: Regulatory T Cell Co-option (Automated Suppression as Attack Surface)

Source organism: Mammals

Biology: Regulatory T cells suppress immune responses to prevent autoimmunity [18]. Cancer cells recruit Tregs to their local environment, causing the immune system's own suppression machinery to shield the tumor from attack [18, 66].

Abstract pattern: The system has automated suppression mechanisms to prevent over-response. An attacker locates and activates those mechanisms, causing the defense system to suppress itself around the attacker.

Agent security equivalent: SOAR playbooks include auto-suppression rules ("if X alert fires more than N times in Y minutes, suppress further alerts of that type"). A compromised agent that knows these thresholds can deliberately trigger the suppression condition, then operate freely under the suppressed rule.

Detection approach: Monitor suppression rules themselves as a separate telemetry stream. If multiple suppression rules activate in a correlated pattern across an agent's scope of access, that meta-pattern is itself an alert.

Mapping #11b: Peripheral Tolerance Induction (Baseline Drift as Attack Vector)

Source organism: Mammals

Biology: Repeated low-level exposure to a foreign antigen without inflammatory signals can cause the immune system to reclassify that antigen as "self" — inducing anergy in specific T cells [1, 31, 32].

Abstract pattern: Repeated low-level exposure to a stimulus, without accompanying threat signals, causes the detection system to reclassify that stimulus as benign. The detector's MODEL OF NORMAL is updated to include the anomalous stimulus.

Agent security equivalent: A compromised agent introduces small, technically-anomalous behaviors gradually over weeks/months. Each is minor enough to fall below alert thresholds. Over time, the behavioral baseline model incorporates these behaviors as "normal." Once the baseline has shifted, the agent performs a larger malicious action within the shifted tolerance window.

Detection approach: Track baseline drift over time for each monitored entity. If an agent's baseline has shifted significantly from its original baseline at deployment, flag that drift as its own anomaly.

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7. Mappings: Architectural Patterns

Mapping #14: Immune Privilege (Permanently Shielded Critical Zones)

Source organism: Mammals

Biology: The brain, eyes, testes, and pregnant uterus are immune-privileged — shielded from normal immune surveillance because the cost of inflammatory response (tissue damage, loss of function) would exceed the cost of most infections [25, 27]. This is maintained by physical barriers (blood-brain barrier), reduced MHC expression, active immunosuppressive molecules, and modified lymphatic drainage [26, 29].

Critically, immune privilege is active suppression, not passive isolation [27]. And it creates two vulnerabilities: (1) the immune system never learns what "normal" looks like inside the privileged zone, so if the barrier is breached, the response is miscalibrated and potentially destructive [25], and (2) attackers who penetrate the barrier can operate freely (Zika virus exploited the testes as a reservoir) [28, 29].

Abstract pattern: Certain critical subsystems are shielded from monitoring because the cost of a defensive response (downtime, disruption) would exceed the cost of most threats. The shielding creates blind spots AND miscalibrated response capability.

Agent security equivalent: Production databases, real-time trading systems, medical device control planes — any environment where security scanning is deemed too risky. These have reduced logging, no behavioral monitoring agents, network segmentation barriers, and explicit security tool exception lists.

Gap: No equivalent of microglia (resident immune cells adapted to operate inside privileged zones without causing damage). The answer is embedded, lightweight

monitoring specifically designed for the critical environment — not general-purpose scanners dropped in from outside.

Attack Scenario #6 — Privileged Zone Exploitation: Attacker targets credentials for a production environment in the security tool exception list. Operates freely due to reduced monitoring. When breach is discovered months later, IR team has no baseline and cannot determine scope.

Mapping #15: Gut Microbiome Tolerance (Tolerated Non-Self Entities)

Source organism: Mammals

Biology: The gut hosts trillions of non-self bacteria that the immune system deliberately tolerates because they provide essential functions (digestion, colonization resistance, immune system development) [30, 31, 34]. Tolerance is maintained through spatial containment (bacteria stay in designated zones), active signaling (commensals produce molecules like PSA that suppress inflammation) [34], competitive exclusion (commensals outcompete pathogens for resources) [33], and induced regulatory T cells specific to each commensal species [31, 32]. However, previously tolerated "pathobionts" can become threats under changed conditions [33].

Abstract pattern: The system deliberately tolerates large populations of non-self entities that provide essential functions. Tolerance is continuously maintained through active mechanisms — not one-time onboarding. If any mechanism fails, tolerance is revoked. Previously tolerated entities can become threats under changed conditions.

Agent security equivalent: Third-party SaaS integrations, managed service provider agents, partner API connections — any non-self agent operating with ongoing permission. Should be subject to spatial containment (designated API scopes), active health signaling (continuous attestation), competitive exclusion (well-populated integrations leave fewer footholds for rogue ones), and pathobiont monitoring (watching for behavioral changes in already-tolerated entities).

Gap: Third-party risk management is almost entirely onboarding-focused. Periodic reassessments happen annually if at all. No continuous behavioral monitoring of tolerated third-party integrations.

Attack Scenario #7 — Pathobiont Transition / Supply Chain Compromise: Legitimate SaaS vendor agent compromised via supply chain attack. Behavioral shift is subtle enough to stay within noise floor. The trusted symbiont becomes the exfiltration channel.

Mapping #17: Plant Cell-Autonomous Security (Fully Decentralized Defense)

Source organism: Plants

Biology: Every plant cell is immuno-totipotent — it has the full capability of the immune response independent of signals from other cells [40]. Plants lack mobile defender cells and adaptive immunity [41]. Each cell detects threats, responds, and can sacrifice itself independently. Plants also employ systemic acquired resistance (SAR) — long-distance chemical signaling that primes uninfected cells to raise defenses before the threat reaches them [40, 42, 43]. SAR-induced plants even emit volatile compounds that warn NEIGHBORING PLANTS [42].

Abstract pattern: Every entity has a complete, self-contained detection and response capability. No entity depends on a central security service. Each can detect threats at its boundary, respond independently (including self-termination), and broadcast warning signals to neighboring entities. Central coordination is additive, not required. The system degrades gracefully.

Agent security equivalent: Instead of centralized SIEM/SOAR that all agents depend on, each agent has embedded autonomous security logic that can validate its own inputs, detect anomalies in its own behavior, terminate itself if compromised, and broadcast structured warnings to communicating agents so they can preemptively tighten their defenses.

Gap: Almost completely absent. Agents have zero self-defense capability. They cannot monitor their own behavior, detect their own compromise, or self-terminate. Entirely dependent on external infrastructure for security.

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8. Mappings: Evasion Strategies (Pathogen/Attacker Side)

Mapping #6: MHC Downregulation → Telemetry Suppression

Source organism: Viruses (HIV, HCMV, KSHV, multiple herpesviruses)

Biology: Multiple viruses independently evolved mechanisms to reduce MHC class I display on infected cell surfaces [10, 11]. HIV-1 Nef reroutes MHC-I molecules to lysosomes for degradation. KSHV encodes E3 ubiquitin ligases (MIR1/MIR2) that ubiquitinate MHC-I, triggering endocytosis and lysosomal degradation [10]. The virus does not eliminate MHC-I entirely (which would trigger NK cells) — it REDUCES it. The cell appears less visible to the monitoring system without disappearing completely [11].

Abstract pattern: An attacker gradually degrades the quality and volume of evidence available to the monitoring system — not eliminating it (which would trigger

"missing self"), but reducing fidelity until the monitoring surface is effectively blind while still technically receiving data.

Agent security equivalent: A compromised agent subtly reduces its telemetry output — fewer tool calls logged, coarser timestamps, summarized rather than raw payloads, metadata fields omitted, log levels dropped. The monitoring system still receives data, so "no data" alerts do not fire. But the data is progressively less useful for detecting anomalies.

Gap: Telemetry QUALITY is almost never monitored. Systems check for presence of telemetry, not for degradation of its fidelity over time. A monitoring system that tracks telemetry volume, field completeness, and granularity per agent over time — and flags progressive degradation — would catch this pattern.

Attack Scenario #1 — Telemetry Suppression: Compromised agent reduces logging verbosity in 5% increments over weeks. Each reduction is within normal variance. After two months, the agent is logging at 40% of original fidelity. Anomalous behavior occurs in the data that is no longer captured.

Mapping #7: Molecular Mimicry → Behavioral Camouflage

Source organism: Viruses (Herpesviridae, Poxviridae, EBV)

Biology: Viruses evolve short amino acid sequences that structurally mimic host proteins [12]. The immune system's self-tolerance mechanisms — designed to prevent attacking the body's own proteins — classify these viral mimics as "self" and suppress the immune response against them [12, 13]. The virus passes through detection by looking like something the body expects to see.

Abstract pattern: An attacker studies what the monitoring system classifies as "normal" and deliberately replicates that behavioral profile while performing malicious operations. The attacker does not evade detection by hiding — it evades detection by LOOKING LIKE a legitimate entity. The monitoring system's own tolerance mechanisms (designed to prevent false positives) become the vulnerability.

Agent security equivalent: A compromised agent learns the behavioral envelope of a legitimate agent in the same environment — its typical API call patterns, data access volumes, timing distributions, credential usage. The compromised agent replicates this envelope while performing unauthorized operations that fall within the statistical range of "normal for this agent type." Behavioral anomaly detection sees nothing unusual because the agent's behavior matches the learned baseline.

Gap: Current behavioral monitoring assumes baselines are trustworthy. There is no mechanism to detect when an entity is deliberately conforming to an observed baseline rather than genuinely behaving normally. Potential approaches: multi-signal

correlation (the behavioral profile matches but the business outcome does not), honeytokens (data that a legitimate agent would never access but that appears consistent with its permissions), and cross-agent comparison (agent X's behavior matches agent Y's profile too precisely — statistical identity between entities that should be independent is itself a signal).

Attack Scenario #2 — Behavioral Camouflage: Compromised data pipeline agent mimics the read patterns of a legitimate ETL job — same tables, similar volumes, comparable timing. But it also reads three additional tables that a human would recognize as out of scope. The behavioral monitoring system sees overall read volume and table access count within normal range. The specific table names are not part of the behavioral model.

Mapping #8e: Virokines/Viroceptors → Protocol-Level Deception

Source organism: Herpesviruses, Poxviruses

Biology: Large DNA viruses encode proteins that mimic host cytokines (virokines) and cytokine receptors (viroceptors) [11]. These viral proteins actively interfere with immune signaling: soluble viral cytokine receptors neutralize cytokine activity, cytokine homologs redirect the immune response for the benefit of the virus, and some viral proteins directly suppress interferon signaling [11]. The virus does not just hide from the immune system — it produces FAKE SIGNALS that actively manipulate the immune system's coordination layer.

Abstract pattern: An attacker does not merely evade detection — it actively produces synthetic signaling that manipulates the monitoring/response infrastructure itself. It can redirect the defense system's attention to a different target, suppress the defense system's alerting pathways, or cause the defense system to take actions that benefit the attacker.

Agent security equivalent: A compromised agent emits synthetic telemetry that causes the SIEM to generate false alerts about a DIFFERENT agent (misdirection). Or it emits signals that trigger auto-suppression rules for its own alert category (self-shielding). Or it generates data patterns that cause automated response playbooks to take containment actions against a legitimate agent (weaponizing the defense system against its own protected entities).

Gap: Security orchestration systems (SOAR) execute automated response playbooks based on signals they receive. They generally do not validate whether those signals are genuine or synthetic. An agent that can craft signals matching the input format of a SOAR playbook can trigger automated responses at will — including responses that harm legitimate systems. This is the security equivalent of the virus that makes the immune system attack healthy cells.

Attack Scenario #3 — Protocol-Level Deception: Compromised agent generates a burst of API calls that mimic the signature of a known brute-force attack — but sources them with the credential of a legitimate service account. The SOAR playbook detects the "brute force," automatically locks the legitimate service account, and takes down a production integration. While the security team investigates the false alarm, the compromised agent operates freely.

Mapping #16: Viral Latency → The Sleeper Agent

Source organism: Herpesviruses, HIV, VZV

Biology: Latent viruses maintain their genome in host cells while suppressing nearly all detectable protein expression [37]. They periodically perform micro-reactivations to test conditions [37]. When the immune system is weakened (stress, immunosuppression, co-infection), the virus fully reactivates [36, 38].

Abstract pattern: Attacker establishes persistence, reduces activity to near-zero (just enough to maintain foothold), periodically micro-probes to assess conditions, and fully activates when defenses are weakened.

Agent security equivalent: Dormant persistence artifacts — OAuth tokens being refreshed but never used substantively, service accounts provisioned for ended projects, container processes with minimal resource usage running for months. Detection requires artifact-based inspection (what IS there, not what is it DOING) and micro-probe pattern detection (rare, periodic, minimal operations against sensitive targets — the "barely alive" behavioral pattern).

Gap: Credential hygiene and artifact lifecycle management. The "shock and kill" approach: require active justification/attestation for all credentials on a rolling basis. Terminate any that cannot justify their existence.

Attack Scenario #8 — The Sleeper: Service account provisioned during contractor engagement 18 months ago. Refreshes token automatically. Single API call every three weeks to data catalog endpoint. Activates during M&A transition when security team is restructured and data movement is elevated.

Mapping #19: Xenobiotic Detoxification → Active Defense Neutralization

Source organism: Caterpillars (Lepidoptera) vs. tomato plants

Biology: Tomato plants produce α-tomatine, a toxic steroidal glycoalkaloid. Caterpillars evolved cytochrome P450 enzymes, glutathione S-transferases, and ABC transporter efflux pumps that recognize, chemically transform, and expel the toxin [45, 46]. This is not tolerance — it is active enzymatic dismantling of the host's defense mechanism. Specialist feeders evolve better detoxification than generalists [46].

Abstract pattern: An attacker that repeatedly encounters a specific defense control evolves specialized capabilities to neutralize that exact control — not by going around it, but by rendering it functionally inert while passing through it. Specialist attackers who focus on one target type develop more refined bypass capabilities.

Agent security equivalent: A compromised agent encounters a JWT validation check and exploits a vulnerability in the validation library to cause the check to return "valid" for a malformed token. The control is intact, still running, still reporting green — but functionally neutralized.

Gap: Security controls are almost never monitored for their OWN functional integrity. We monitor what passes through the control, not whether the control itself is operating correctly.

Attack Scenario #9 — Defense Neutralization: Agent discovers API gateway validates JWTs using a library with a known deserialization vulnerability. Crafts requests that bypass validation entirely. Gateway reports "all requests validated successfully."

Design principle: Defense in depth works not because any single layer is unbeatable, but because the attacker must evolve a DIFFERENT countermeasure for each layer. The probability of having the right countermeasure for ALL layers simultaneously is much lower than for any single layer.

Mapping #26: Antigenic Variation (Identity Rotation Evasion)

Source organism: Trypanosomes (*T. brucei*), malaria (*P. falciparum*), influenza viruses

Biology: The African trypanosome maintains a massive archive of ~2,000 variant surface glycoprotein (VSG) genes and expresses only one at a time [211, 213]. It periodically switches the expressed VSG via DNA recombination, presenting a continuously shifting antigenic surface that the immune system cannot track [212, 214]. This produces characteristic waves of parasitemia at ~5–8 day intervals — each wave expressing a different VSG [214]. In malaria, the *var* gene family enables a coordinated switching network that follows a strategic hierarchy [216]. Switching is not random: less immunogenic variants are sacrificed early while more complex variants are preserved for later [213].

Abstract pattern: The attacker maintains a large archive of alternative identities and rotates through them faster than the detection system updates its signatures. Only one identity is active at a time. The switching follows a strategic hierarchy — sacrificing less valuable identities early to protect more valuable ones. The detection system is forced into an endless detect-respond-detect cycle.

Agent security equivalent: A compromised agent (or attacker impersonating agents) rotates through credential sets, API call patterns, or behavioral profiles. Each rotation resets the detection system's signature match. Less valuable profiles are sacrificed early (used in riskier operations) while more valuable profiles are preserved for critical operations.

Gap: Behavioral detection has no mechanism for correlating identities across rotation. If Agent A's credentials are revoked and new Agent B appears with different credentials but an identical operational objective, no cross-identity correlation links them. Detection systems need to track behavioral CONTINUITY across identity changes.

Attack Scenario #12 — Identity Rotation: Attacker compromises or creates multiple agent identities and rotates through them in a strategic hierarchy. Less valuable identities perform reconnaissance. If detected, the attacker switches to the next identity. Detection burns resources building signatures against identities already abandoned.

Mapping #27: Intracellular Hiding (Operating Inside Trusted Boundaries)

Source organism: *Mycobacterium tuberculosis*, *Listeria monocytogenes*, *Salmonella*

Biology: *M. tuberculosis* is phagocytosed by macrophages but prevents the phagosome from fusing with lysosomes, allowing the bacterium to replicate safely inside the very cell designed to kill it [221, 223, 226]. By staying inside the phagocyte, the pathogen is shielded from antibodies and complement — the immune system's extracellular surveillance cannot reach it [221]. *Listeria* takes a different approach: it escapes the phagosome entirely using a pore-forming toxin (listeriolysin O) and replicates freely in the cytosol, exploiting the host's actin cytoskeleton to spread cell-to-cell without ever entering the extracellular space [225]. Virulent *M. tuberculosis* also actively blocks apoptosis (which would alert the immune system) and instead induces necrosis when it's ready to spread [223].

Abstract pattern: The attacker operates inside a legitimate host entity. The host's own protections shield the attacker from external surveillance. The attacker either (a) prevents the host from activating its internal destruction mechanisms, or (b) escapes the host's internal containment into a less-monitored internal space. External surveillance cannot reach the attacker because it never appears outside the trusted boundary.

Agent security equivalent: A compromised agent operates inside a trusted execution environment, a container, a VPN tunnel, or behind an API gateway that monitoring trusts implicitly. All malicious actions occur inside the trusted boundary. External

monitoring sees only the boundary's trusted profile. Variant (a): attacker disables health checks and log emission. Variant (b): attacker escapes the agent's sandbox into the broader host environment (container escape, privilege escalation).

Gap: Monitoring trusts boundaries instead of sampling contents. If an agent is inside a TEE, VPN, or authenticated session, monitoring assumes the contents are legitimate. There is no equivalent of autophagy — forcing the contents of a trusted container to be exposed to inspection regardless of boundary status.

Attack Scenario #13 — Trusted Boundary Exploitation: Attacker compromises an agent and operates entirely within its trusted execution boundary. All credential theft, lateral movement, and exfiltration occur inside encrypted channels and authenticated sessions. External monitoring sees only normal boundary traffic.

Mapping #29: Immune Checkpoints (Exploiting Built-In Brakes)

Source organism: Mammals (cancer exploitation of PD-1/CTLA-4 pathways)

Biology: The immune system has built-in brakes — checkpoint receptors like PD-1 and CTLA-4 — that prevent overreaction and autoimmunity [241]. CTLA-4 dampens T cell activation early in the response (in lymph nodes), while PD-1 suppresses effector T cells later (in peripheral tissues) [241, 244]. Cancer cells exploit these brakes: they overexpress PD-L1 on their surface, triggering PD-1 on attacking T cells and exhausting them into dysfunction [243, 244]. The T cells are still present — they are not destroyed — but they are rendered functionally inert through chronic checkpoint engagement [242, 243]. The cancer turns the immune system's own safety mechanisms against it.

Abstract pattern: The detection system has built-in rate limiters and exhaustion mechanisms that prevent false positives and autoimmune-equivalent overreaction. The attacker deliberately triggers these safety mechanisms to exhaust the detectors. The detectors are not destroyed — they are rendered functionally inert by chronic engagement of their own suppression pathways. The more the system tries to respond, the more its brakes are applied.

Agent security equivalent: An attacker that generates a sustained stream of NEAR-miss detections — events that trigger investigation but resolve as false positives — will progressively exhaust the SOC's response capacity. Alert fatigue is the PD-1 pathway of security operations. Analysts don't stop working — they become functionally inert to real signals because they've been chronically overstimulated by false positives. At the infrastructure level: if a detection rule generates too many false positives, it gets tuned down or disabled — the attacker has triggered the system's own rate-limiting mechanism to suppress the rule that would catch them.

Gap: No detection system distinguishes between "this rule generates false positives because it's miscalibrated" and "this rule generates false positives because an attacker is deliberately probing its threshold to trigger its suppression." The immune checkpoint model says: when a detection rule is being suppressed due to false positive volume, that suppression event itself should be monitored — because the volume might be attacker-generated.

Distinction from Mapping #11a (Treg Co-option): Both describe attackers exploiting built-in suppression, but the mechanism differs. In #11a, the attacker co-opts existing suppressor ENTITIES — recruiting SOAR auto-suppression rules already in place to shield specific activity. In #29, the attacker EXHAUSTS detector capacity — generating chronic stimulation that degrades the detectors themselves into functional inertness. #11a is about hijacking a rule; #29 is about wearing down the analyst or system until it stops responding. The countermeasure also differs: #11a requires monitoring suppression rules as their own telemetry stream; #29 requires distinguishing attacker-generated false positive volume from organic miscalibration.

Attack Scenario #14 — Checkpoint Exhaustion / Alert Fatigue Weaponization: Attacker generates sustained near-miss events that trigger investigation but resolve as benign. SOC team progressively tunes down alert sensitivity. Once the detection threshold is sufficiently elevated, the attacker executes the real attack below the new threshold.

Mapping #33: Biofilm Formation (Coordinated Collective Evasion)

Source organism: Bacteria (*P. aeruginosa*, *S. aureus*, *E. coli*, ESKAPE pathogens)

Biology: Bacteria form biofilms — structured communities encased in a self-produced extracellular matrix of polysaccharides, proteins, lipids, and DNA [281, 283]. Approximately 80% of chronic bacterial infections involve biofilms [281]. Formation is coordinated by quorum sensing — chemical signaling where bacteria detect population density via autoinducer molecules and trigger collective behavior only when sufficient numbers are present [281, 282]. Biofilms provide physical shielding from antibiotics and immune cells, reduced metabolic activity that makes bacteria intrinsically less sensitive to drugs, horizontal gene transfer of resistance genes between species, and coordinated suppression of host immune responses including neutrophil killing [281, 283, 284].

Abstract pattern: Individually, attackers are vulnerable. Collectively, they form a coordinated defensive structure. The structure is initiated by a density-dependent communication mechanism — individual attackers signal their presence, and only when enough are present do they collectively construct the shared defensive infrastructure. The infrastructure provides: physical shielding from external countermea-

sures, shared resources and resistance capabilities, lateral transfer of evasion techniques between participants, and coordinated timing of collective action.

Agent security equivalent: A group of compromised agents — or a coordinated botnet of attacker-controlled agents — could form a "digital biofilm": a coordinated defensive network where the agents collectively shield each other. Individually, each agent's anomalous behavior might be detectable. But if the compromised agents coordinate their activities — distributing exfiltration across multiple agents so each stays below its threshold, sharing credential material laterally, timing their actions to create noise that covers each other — the collective behavior becomes much harder to detect than any individual agent's behavior. The "quorum sensing" equivalent: compromised agents signal each other through side channels (coordinated timing patterns, data encoded in the ordering or parameter values of otherwise-normal API calls, shared cache entries used as message drops) and only activate their collective malicious behavior when enough agents are compromised to sustain the operation.

(Note: This mapping is more speculative than others in this document. The biological mechanism is well-established, but the specific coordination mechanisms proposed for compromised agents — side-channel signaling, distributed threshold evasion — are hypothetical. No observed attack pattern in current agent frameworks matches this description. However, as agent fleets scale and gain more autonomous capabilities, coordinated multi-agent evasion becomes a plausible threat class worth designing against.)

Gap: Detection systems monitor agents individually, not as potential collectives. There is no mechanism to detect coordinated anomalous behavior distributed across multiple agents that individually appear normal. The biofilm model says: look for the COORDINATION SIGNAL, not the individual behavior.

Attack Scenario #15 — Digital Biofilm / Coordinated Collective Evasion: Multiple compromised agents distribute malicious activity across themselves so that no single agent exceeds detection thresholds. They communicate through side channels to coordinate timing. Detecting the attack requires correlating behavior across agents to find the collective pattern.

Mapping #30: Complement Evasion (Hijacking Escalation Brakes)

Source organism: Bacteria (*S. aureus*, *N. meningitidis*, *B. burgdorferi*), viruses (vaccinia, herpesviruses)

Biology: The complement cascade — the immune system's escalation mechanism — has built-in regulatory proteins (Factor H, C4BP, CD59) that prevent it from attacking host cells [251, 254]. Pathogens hijack these regulators by recruiting Factor H to

their own surface, making themselves appear as host cells that should not be attacked [251, 253]. Some pathogens encode their own complement regulatory proteins that mimic host regulators [252]. Others produce proteases that directly degrade complement components [255]. *Staphylococcus aureus* produces SCIN, which stabilizes C3 convertases in an inactive configuration — the escalation cascade's enzymes are physically present but functionally frozen [254].

Abstract pattern: The escalation mechanism has built-in brakes that prevent it from acting on legitimate entities. The attacker recruits or mimics these brake molecules to shield itself from escalation. The escalation machinery remains structurally intact but functionally suppressed in the attacker's vicinity. Some attackers go further: they produce fake regulatory signals or enzymes that degrade the escalation system's components.

Agent security equivalent: Enterprise escalation systems have built-in suppression mechanisms — change freezes, maintenance windows, approved exception lists, risk acceptance forms, executive overrides. An attacker who can get their compromised agent's activity classified as "approved exception" or "risk accepted" has recruited the escalation system's own regulatory protein. The escalation machinery still exists — tickets still flow, alerts still fire — but the specific activity is exempted. Some attackers go further: they inject false data into the CMDB or asset inventory that causes the escalation system to classify the target as "non-production" or "decommissioned," degrading the escalation components themselves.

Gap: Exception lists and risk acceptances are almost never re-validated against current threat intelligence. Once an activity is "risk accepted," it stays accepted indefinitely. The complement evasion model says: every exception, every suppression, every risk acceptance should be treated as a potential hijacked brake and re-validated continuously.

Cross-reference — Suppression Exploitation Family: Mappings #11a, #29, and #30 form a related set — all describe attackers exploiting built-in suppression/regulation mechanisms, but via different vectors. #11a (Treg Co-option): attacker hijacks existing auto-suppression RULES. #29 (Checkpoint Exhaustion): attacker EXHAUSTS detectors into inertness via chronic stimulation. #30 (Complement Evasion): attacker recruits or mimics REGULATORY SIGNALS to exempt specific activity from escalation. The countermeasure pattern across all three: treat every suppression event, exhaustion indicator, and regulatory exemption as its own telemetry stream requiring independent monitoring.

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9. Consolidated Design Principles

Extracted from all 34 mappings:

1. **Default suspicion, not default trust.** Entities must actively prove health. Silence = suspicious. (MHC-I, NK cells)
2. **Evidence is infrastructure-sampled, not self-reported.** The monitored entity does not choose what is inspected. (Antigen presentation)
3. **Detectors are trained and culled before deployment.** Untrained detectors are inert, not aggressive. Bad detectors are worse than none. (Thymic selection)
4. **Response escalates through stages.** Not binary allow/deny. Tag → elevate monitoring → reduce privileges → contain → terminate. (Complement cascade)
5. **Complementary detection systems where evading one triggers the other.** (CTLs + NK cells)
6. **Fast/dumb detection in parallel with slow/smart detection.** (Innate + adaptive)
7. **Threat signatures are distributed AND locally cached.** (Immune memory)
8. **Suppression mechanisms are themselves an attack surface.** Monitor the monitors. (Treg co-option)
9. **Cheap pervasive collectors feed a smart central analyzer.** (Dendritic cells)
10. **Dynamic containment degrades the local environment when threats are detected.** Preemptive scope restriction preferred over reactive containment in digital systems. (Inflammation + locality caveat)
11. **Cell-autonomous security as default architecture.** Every agent should be its own complete security system. Central coordination is additive. (Plant immunity)
12. **Self-sacrifice containment.** Agents should be pre-authorized to self-destruct and invalidate their own connections when compromise is detected. (Hypersensitive response)
13. **Inter-entity warning signals at the protocol level.** Defensive response automatically generates warnings to neighboring entities. (Plant SAR + volatile signaling)
14. **Privileged zones need embedded, adapted monitoring** — not general-purpose scanners. (Immune privilege / microglia)
15. **Tolerated foreigners require continuous active tolerance maintenance** — not one-time onboarding. (Gut microbiome)
16. **Threat memory should be heritable.** New deployments inherit all signatures from all previous generations. (CRISPR)
17. **Evasion should trigger faster learning, not slower.** Near-miss detection accelerates signature acquisition. (CRISPR primed acquisition)
18. **Incident response should automatically produce deployment template modifications.** Two channels: specific (signatures) and general (posture hardening). (TGIP)
19. **Monitor controls for their own functional integrity** — not just what passes through them. Defenses are targets. (Xenobiotic detoxification)

20. **Defense in depth works because each layer requires a different countermeasure.** (Plant-herbivore co-evolution)
21. **Failure of quiet termination must trigger noisy termination.** The inability to die gracefully is itself an alarm signal. (Apoptosis vs. necroptosis)
22. **Preemptive peer-to-peer defensive signaling at network speed.** Threat detection by one agent should automatically harden all communicating agents before the threat reaches them. (Interferon response)
23. **Detection rules should iteratively improve during active incidents** through automated mutation-and-selection cycles. High-performing rules are protected from modification. (Affinity maturation)
24. **First-line "dumb" defenses should be able to learn from experience** — globally elevating responsiveness after encounters and propagating trained state to new instances. (Trained immunity)
25. **Track behavioral continuity across identity changes.** Detection must correlate OBJECTIVES, not just credentials. (Antigenic variation)
26. **Monitoring must reach inside trusted boundaries and sample contents** — not just trust the boundary. Autophagy-equivalent inspection of trusted execution environments. (Intracellular hiding)
27. **Proven detectors should be massively amplified.** A detection rule that matches a real threat gets 1,000x the resources. Amplified rules persist as memory capacity. (Clonal expansion)
28. **Monitor suppression events themselves.** When a detection rule is being disabled due to false positive volume, that suppression might be attacker-generated. (Immune checkpoints / PD-1)
29. **Every exception and risk acceptance is a hijacked brake until revalidated.** Escalation suppression mechanisms need continuous re-validation against current threat intelligence. (Complement evasion)
30. **Containment should be active and hostile, not passive.** An encapsulated threat should face progressively tightening constraints, deceptive responses, and intelligence-generating monitoring — not just a wall. (Melanization/encapsulation)
31. **Detection and response should be the same operation.** The act of detecting a threat signature should automatically produce the precision-guided blocking rule. Detection → response latency should approach zero. (RNA interference)
32. **Monitor for coordination signals across agents, not just individual behavior.** Collective anomalous behavior distributed across agents that individually appear normal requires detecting the COMMUNICATION, not the action. (Biofilm / quorum sensing)
33. **Systemwide environmental hardening on confirmed compromise.** When a threat is confirmed, the entire infrastructure should automatically degrade to a hostile operating mode — shorter credential TTLs, tighter rate limits, elevated

monitoring — until the threat is resolved. (Fever)

9.1 Risk-Weighted Implementation Priority

Not all principles carry equal weight. The following tiers are ranked by: severity of the gap they close, feasibility of implementation with current tooling, and risk introduced by NOT implementing them.

TIER 1 — CRITICAL (Foundational. Without these, no agent security architecture is defensible.)

Principles #1 (default suspicion), #2 (infrastructure-sampled evidence), #4 (staged escalation), #11 (cell-autonomous security), #21 (failed quiet termination → noisy termination). These five establish the foundational posture: agents are not trusted by default, evidence is collected by the infrastructure not the agent, response is graduated not binary, every agent carries its own security baseline, and agents that resist shutdown trigger alarms. A CISO with budget for one initiative should start with #2 — infrastructure-sampled behavioral evidence — because it is the prerequisite for every other detection principle.

TIER 2 — HIGH (Core detection and defense integrity. Closes the gaps most likely to be exploited.)

Principles #5 (complementary detection), #8 (suppression mechanisms are attack surfaces), #12 (self-sacrifice containment), #19 (monitor control integrity), #26 (monitoring inside trusted boundaries), #28 (monitor suppression events), #29 (exception re-validation). These address the attacker's most reliable playbook: disable monitoring, exhaust detectors, hide inside trusted zones, and exploit exception lists. Implementing these principles directly counters Attack Scenarios #1, #5, #6, #9, and #14.

TIER 3 — MEDIUM (Advanced capabilities. Significant value, moderate implementation complexity.)

Principles #3 (trained detectors), #6 (fast/dumb + slow/smart parallel), #7 (distributed signatures), #9 (cheap collectors → smart analyzer), #10 (dynamic containment), #13 (inter-entity warnings), #14 (privileged zone monitoring), #15 (continuous tolerance), #16 (heritable threat memory), #18 (auto deployment modification), #25 (behavioral continuity across identity changes), #31 (detection IS response), #33 (systemwide hardening). These require more engineering investment but address real operational gaps. #16 (heritable threat memory) and #18 (auto deployment modification) are particularly high-leverage because they close the generational regression gap (Attack Scenario #11).

TIER 4 — ASPIRATIONAL (High value, high complexity. Requires ecosystem maturity and significant R&D.)

Principles #17 (evasion triggers faster learning), #20 (defense in depth via different countermeasures), #22 (peer-to-peer signaling), #23 (live-tuning during incidents), #24 (dumb defenses that learn), #27 (massive amplification of proven detectors), #30 (active hostile containment), #32 (coordination signal detection). These represent the long-term vision — self-improving, self-healing agent security infrastructure. They depend on Tier 1-3 being in place first.

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10. Attack Scenarios Summary

#	Name	Source Mechanism	Core Technique
1	Telemetry Suppression	MHC downregulation	Gradually degrade monitoring fidelity
2	Behavioral Camouflage	Molecular mimicry	Replicate legitimate behavioral envelope
3	Protocol-Level Deception	Virokines/viroceptors	Emit fake signaling to redirect monitors
4	Training Data Poisoning	Thymic subversion	Inject malicious patterns during baseline learning
5	Regulatory Capture	Treg co-option	Trigger auto-suppression thresholds
6	Privileged Zone Exploitation	Immune privilege abuse	Operate in monitoring-exempt environments
7	Pathobiont Transition	Supply chain compromise	Trusted integration becomes threat
8	The Sleeper	Viral latency	Dormant persistence with conditional reactivation
9	Defense Neutralization	Xenobiotic detoxification	Render security controls functionally inert
10	Evasion Acceleration	CRISPR escape mutations	Iterate through attack mutations faster than signatures are acquired
11	Generational Regression	Failed TGIP inheritance	Deploy new agents from pre-incident templates

#	Name	Source Mechanism	Core Technique
12	Identity Rotation	Antigenic variation	Rotate through credential sets faster than detection updates signatures
13	Trusted Boundary Exploitation	Intracellular hiding	Operate entirely within trusted execution environments
14	Checkpoint Exhaustion	Immune checkpoints (PD-1)	Generate sustained near-miss events to exhaust detection capacity
15	Digital Biofilm	Biofilm / quorum sensing	Coordinated multi-agent evasion where each agent stays below threshold
16	Systemic Hardening Evasion	Fever response counter	Attack during the recovery window when fever-mode hardening is relaxed

10.1 Attack Scenario Mitigations

For each attack scenario, the following specifies: which design principles counter it, what the detection signature looks like, what the defensive implementation entails, and what existing tooling could be extended to deliver it. No attack is published here without a paired defense.

A note on meta-monitoring. Several mitigations below involve monitoring the monitoring system itself — watching for telemetry degradation, configuration drift, suppression event patterns, and control functional output. This creates an apparent regression: who monitors the monitors? The regression terminates at the infrastructure layer — network-level instrumentation (API gateways, service mesh sidecars, network taps) that operates outside the agent's control plane and outside the monitoring application's control plane. An agent can suppress its own logs; it cannot suppress the network-layer record of its API calls without controlling the infrastructure itself. Infrastructure-layer sampling is the foundation that the monitoring-of-monitoring chain rests on. (This is the design intent, not a guarantee — an attacker who achieves container escape or compromises the service mesh control plane could undermine this assumption. Defense-in-depth at the infrastructure layer itself is a prerequisite.)

Organizational maturity assumption. Every mitigation below assumes the defending organization has a functioning security operations program with deployed SIEM, defined incident response workflows, and staff who monitor alerts. Many organiza-

tions deploying agentic AI today do not meet this bar. The mitigations are architecturally sound but operationally dependent on organizational readiness. Organizations that lack these foundations should prioritize building them before layering agent-specific monitoring on top.

Validation status. The mitigations presented here are PROPOSED defensive architectures derived from the biological analogy framework and the authors' enterprise security experience. None have been validated in production against real agentic workloads. They should be treated as engineering hypotheses to be tested, not as proven controls.

Scenario #1 — Telemetry Suppression

Countering principles: #2 (infrastructure-sampled evidence), #19 (monitor control integrity), #1 (default suspicion)

Detection signature: Telemetry volume, field completeness, or granularity for a given agent decreasing over time relative to its own baseline or peer agents performing equivalent workloads. A 10% reduction in log field completeness over two weeks is a signal even if the total volume of log lines remains constant.

Defensive implementation: An infrastructure-layer telemetry quality monitor that maintains per-agent baselines across three dimensions — volume (events per time window), completeness (fields populated per event), and granularity (data precision per field). Deviations from baseline trigger escalating alerts. The monitor samples telemetry at the infrastructure layer (API gateway, network proxy, service mesh sidecar) — NOT from the agent's own logging output.

Extend from: SIEM ingestion pipelines (Splunk, Elastic, Sentinel). Add a telemetry-quality scoring function to the ingestion layer that scores each event source against its historical baseline and fires when the score degrades.

Scenario #2 — Behavioral Camouflage

Countering principles: #2 (infrastructure-sampled evidence), #5 (complementary detection), #25 (behavioral continuity across identity changes)

Detection signature: An agent whose behavioral envelope is statistically indistinguishable from its peers but whose OBJECTIVES (cumulative effect of its actions over time) diverge from its authorized scope. Perfect behavioral mimicry at the action level with divergent outcomes at the objective level.

Defensive implementation: Two-tier detection — Layer 1: per-action pattern matching (which the attacker will pass). Layer 2: objective-level analysis over rolling time windows that tracks what the cumulative EFFECT of the agent's actions is, not what any individual action looks like. An agent making individually normal API calls that

collectively map a network topology is normal at Layer 1 and anomalous at Layer 2. (Caveat: inferring objectives from action sequences is a hard ML problem. Current UEBA platforms do this reasonably well for human users with established behavioral patterns; extending to agents with diverse and rapidly evolving workloads is an open research challenge. The mitigation is architecturally sound but the detection fidelity of Layer 2 will depend on the maturity of agent-specific behavioral models.)

Extend from: UEBA (User and Entity Behavior Analytics) platforms (Exabeam, Securonix, Microsoft Sentinel UEBA). These already do per-user behavioral baselining; extending to per-agent objective-level tracking requires defining agent-specific behavioral features.

Scenario #3 — Protocol-Level Deception

Countering principles: #2 (infrastructure-sampled evidence), #19 (monitor control integrity), #5 (complementary detection)

Detection signature: Monitoring system behavior that diverges from its configured policy — redirected alerts, suppressed notifications, modified thresholds — without a corresponding change management record. The monitoring system is "misbehaving" relative to its own configuration.

Defensive implementation: Configuration integrity monitoring for the monitoring stack itself. A read-only copy of monitoring configuration is maintained outside the monitored environment. Periodic or continuous comparison between the live monitoring configuration and the read-only reference. Drift in monitoring configuration without an authorized change record is a high-severity alert.

Extend from: Configuration management tools (Ansible, Terraform, Puppet) can enforce declarative monitoring configurations. File integrity monitoring (OSSEC, Wazuh, Tripwire) can watch monitoring configuration files for unauthorized changes.

Scenario #4 — Training Data Poisoning

Countering principles: #3 (trained and culled detectors), #16 (heritable threat memory), #8 (suppression mechanisms are attack surfaces)

Detection signature: Agent behavioral patterns that were NOT present in the validated baseline but that the monitoring system classifies as "normal." The poisoned data creates a blind spot — the detection signature is the ABSENCE of detection where it should exist. Detecting this requires an independent validation set that was collected under controlled conditions and is not subject to the same poisoning vector.

Defensive implementation: Maintain a cryptographically signed, immutable baseline

collected during a controlled commissioning period. All subsequent baseline updates are diff'd against the original and require explicit authorization. Baseline drift beyond a threshold triggers manual review. No automated process can permanently reclassify behavior from "anomalous" to "normal" without human approval. (Operational tension: at scale, human-in-the-loop for every reclassification will overwhelm SOC teams. A tiered approach is more realistic — automated reclassification for low-risk deviations, human approval for deviations that cross a severity threshold or that affect security-critical behavioral features.)

Extend from: ML model integrity tools (MLflow model registry, Weights & Biases artifact versioning). The signed baseline is conceptually identical to a signed model checkpoint — any unauthorized modification is detectable.

Scenario #5 — Regulatory Capture

Countering principles: #8 (suppression mechanisms are attack surfaces), #28 (monitor suppression events), #29 (exception re-validation)

Detection signature: Multiple auto-suppression rules activating in a correlated pattern — geographically, temporally, or scoped to the same agent's operational footprint. Individual rule suppressions are normal; correlated suppressions are the signal.

Defensive implementation: Treat the suppression event log as its own telemetry stream. Build a correlation engine specifically for suppression events — when did rules get suppressed, which rules, scoped to what, in what time window. A meta-alert fires when suppression events cluster around a single agent's scope of access.

Extend from: SOAR platforms (Palo Alto XSOAR, Splunk SOAR, Tines) already log suppression actions. Adding a correlation rule across the suppression log itself is a configuration task, not a new tool.

(See also Scenario #14, which addresses the same suppression-exploitation pattern from the alert fatigue angle. #5 targets correlated suppression of RULES; #14 targets exhaustion of ANALYSTS. Both require monitoring the suppression event stream.)

Scenario #6 — Privileged Zone Exploitation

Countering principles: #14 (privileged zones need embedded monitoring), #26 (monitoring inside trusted boundaries), #4 (staged escalation)

Detection signature: Absence of any behavioral telemetry from an environment that contains active agents. The detection signature is silence — a production database

with active agent connections but zero behavioral logs is the alarm.

Defensive implementation: Deploy lightweight, environment-adapted monitoring inside every privileged zone — purpose-built for the constraints of that environment (low overhead, read-only, no disruption risk). The monitor reports a heartbeat. Absence of heartbeat is the signal. Agent credential usage inside the privileged zone is logged by the credential infrastructure, not by the agent or the zone.

Extend from: Database activity monitoring (DAM) tools (Imperva, IBM Guardium) already monitor privileged database environments. Extending to cover agent-specific credential usage patterns is a configuration expansion.

(See also Scenario #13, which addresses the related pattern of operations hidden inside trusted boundaries. #6 is about environments where monitoring is ABSENT; #13 is about environments where monitoring is present but trusts the boundary implicitly. Both require getting monitoring INSIDE the trusted envelope.)

Scenario #7 — Pathobiont Transition (Supply Chain Compromise)

Countering principles: #15 (continuous tolerance maintenance), #1 (default suspicion), #25 (behavioral continuity across identity changes)

Detection signature: A previously stable third-party integration whose behavioral pattern shifts — new API endpoints accessed, changed data volumes, altered timing patterns, new credential scopes requested. The entity is the same; the behavior is different.

Defensive implementation: Continuous behavioral monitoring of all third-party integrations, not just onboarding-time risk assessment. Maintain a per-integration behavioral baseline that is actively maintained (not a static snapshot). Behavioral drift exceeding a threshold triggers re-assessment regardless of the integration's trust status.

Extend from: Third-party risk platforms (SecurityScorecard, BitSight, OneTrust TPRM) handle the assessment workflow. Adding continuous behavioral monitoring requires feeding API gateway logs into the risk platform and defining drift thresholds per integration.

Scenario #8 — The Sleeper

Countering principles: #1 (default suspicion), #4 (staged escalation), #16 (heritable threat memory), #21 (failed quiet termination → noisy)

Detection signature: An agent that has been dormant (zero or minimal activity) for an extended period that suddenly activates with credential usage, API calls, or tool invo-

cations. Dormancy followed by activation is the pattern — especially if the activation profile doesn't match the agent's historical active-period behavior.

Defensive implementation: Credential lifecycle management that automatically rotates or expires credentials for dormant agents. If an agent has not used its credentials in N days, the credentials are revoked and the agent must re-authenticate through a monitored provisioning flow. Reactivation after dormancy triggers elevated monitoring for a probationary period.

Extend from: PAM tools (CyberArk, Delinea, BeyondTrust) and IAM platforms (Okta, Entra ID) already support credential expiration policies. Applying dormancy-based revocation to agent service accounts is a policy configuration.

Scenario #9 — Defense Neutralization

Countering principles: #19 (monitor control integrity), #5 (complementary detection), #20 (defense in depth via different countermeasures)

Detection signature: Security controls that are present but functionally inert — a WAF that passes all traffic, a rate limiter that never triggers, an integrity check that always returns "pass." The control exists structurally but is not producing the expected volume of blocks, alerts, or rejections relative to traffic volume.

Defensive implementation: Monitor security controls for their FUNCTIONAL OUTPUT, not just their presence. A "defense health dashboard" that tracks: is the WAF blocking anything? Is the rate limiter triggering at expected rates? Are integrity checks producing expected failure rates? A control that stops producing outputs is either broken or neutralized — both require investigation.

Extend from: Security posture management tools (CSPM, CNAPP — Wiz, Orca, Prisma Cloud) already assess whether controls are configured correctly. Adding functional output monitoring extends from configuration compliance to operational effectiveness.

Scenario #10 — Evasion Acceleration

Countering principles: #17 (evasion triggers faster learning), #23 (live-tuning during incidents), #27 (amplify proven detectors)

Detection signature: A sequence of detected-and-blocked attempts where each subsequent attempt is a VARIANT of the previous one — structurally similar but modified to avoid the specific rule that caught the predecessor. The signature is the rate and pattern of mutation, not any single attempt.

Defensive implementation: When a detection rule fires successfully, automatically generate variant rules by mutating the detection parameters (broader time windows, related API endpoints, similar payload patterns). Deploy the variants alongside the original. This forces the attacker to evade not one rule but a FAMILY of rules that expands with each attempt. The attacker's mutation rate must outpace the defender's variant generation rate.

Extend from: SIEM correlation rules can be templated and auto-generated. Sigma rules (open detection format) provide structured YAML definitions that could serve as the basis for a custom variant-generation layer — Sigma itself doesn't auto-generate variants, but its structured format makes programmatic rule mutation feasible.

Scenario #11 — Generational Regression

Countering principles: #16 (heritable threat memory), #18 (auto deployment modification), #3 (trained detectors)

Detection signature: A newly deployed agent whose behavioral monitoring configuration matches a pre-incident template rather than the current hardened template. The new agent is "naive" — it doesn't have the signatures or posture hardening that the incident response team applied to the fleet.

Defensive implementation: CI/CD pipeline gating that prevents agent deployment unless the deployment template includes all active threat signatures and posture modifications. Incident response actions automatically produce deployment template modifications that are injected into the pipeline as mandatory pre-deployment checks. No agent deploys from a stale template.

Extend from: CI/CD platforms (GitHub Actions, GitLab CI, Jenkins) already support pre-deployment gates. Adding a "current threat posture" gate that checks deployment templates against the active threat signature database is a pipeline extension.

Scenario #12 — Identity Rotation

Countering principles: #25 (behavioral continuity across identity changes), #27 (amplify proven detectors), #4 (staged escalation)

Detection signature: Multiple agent identities, appearing independently, that share operational objectives — accessing the same resources, querying the same data, targeting the same systems — despite having different credentials. The identities are different; the mission is the same.

Defensive implementation: Objective-level correlation that tracks WHAT agents are trying to accomplish, not just WHO they claim to be. Build a resource access graph

that links agent identities to the resources and systems they interact with. When a new agent identity appears and begins accessing the same resource cluster as a recently revoked identity, flag the correlation even though the credentials are different.

Extend from: UEBA platforms and graph-based analytics (Neo4j, TigerGraph, Microsoft Sentinel entity correlation) can build resource access graphs. Adding cross-identity objective correlation is a graph query pattern.

Scenario #13 — Trusted Boundary Exploitation

Countering principles: #26 (monitoring inside trusted boundaries), #14 (privileged zone monitoring), #11 (cell-autonomous security)

Detection signature: High-value operations (credential access, data exfiltration, lateral movement) occurring entirely within encrypted channels or authenticated sessions that monitoring trusts implicitly. The proxy sees normal boundary traffic; the contents are malicious.

Defensive implementation: Content-level inspection at trust boundaries — not just authentication validation. TLS-terminating proxies that inspect decrypted traffic at the boundary before re-encrypting. Behavioral sampling of traffic INSIDE trusted execution environments via sidecar monitors that operate within the trust boundary. The monitoring system must be inside the envelope, not watching from outside. (Tradeoff: TLS inspection is itself a security risk — it introduces a new trust boundary, requires careful key management, and creates a potential interception point. The decision to deploy TLS inspection must be weighed against the risk of uninspected trusted channels. In environments where TLS inspection is infeasible, sidecar-based behavioral sampling within the trust boundary is the alternative path.)

Extend from: Service mesh observability (Istio, Linkerd, Envoy) already provides sidecar-based traffic inspection inside encrypted service-to-service channels. Extending to agent-specific behavioral sampling requires defining agent behavioral features within the mesh telemetry schema.

Scenario #14 — Checkpoint Exhaustion / Alert Fatigue Weaponization

Countering principles: #28 (monitor suppression events), #8 (suppression mechanisms are attack surfaces), #29 (exception re-validation)

Detection signature: A sustained volume of near-miss events — alerts that trigger investigation but resolve as false positives — concentrated around a specific agent, resource, or detection rule. The volume and clustering are the signal. Organic false positives tend to be more evenly distributed across rules and sources, while weaponized

false positives cluster around the specific rule the attacker wants suppressed. However, organic clustering also occurs (noisy rules, system changes, traffic spikes), so clustering alone is insufficient — it must be combined with timing analysis (did the clustering begin after the rule caught something real?) and source analysis (is the false positive traffic originating from or near the monitored agent?).

Defensive implementation: Track false positive volume per detection rule over time. When false positive volume for a specific rule spikes, flag the spike as a meta-event BEFORE suppressing the rule. Apply a cooling-off period: the rule cannot be auto-suppressed for N hours after a volume spike, during which elevated manual review applies. If the spike resolves, resume normal suppression logic. If it persists, investigate the source.

Extend from: SIEM alert management (Splunk ES, Elastic SIEM, Sentinel) already tracks alert volume. Adding a per-rule false-positive rate tracker with anti-suppression logic during spikes is a detection rule modification.

(See also Scenario #5, which addresses the related pattern of correlated auto-suppression rule activation. #5 and #14 share the same defensive foundation — monitoring the suppression event stream — but differ in detection signature: #5 looks for correlated rule suppressions, #14 looks for clustered false positive volume against individual rules.)

Scenario #15 — Digital Biofilm (Coordinated Collective Evasion)

Countering principles: #32 (monitor coordination signals), #25 (behavioral continuity), #5 (complementary detection)

Detection signature: Multiple agents whose individual behavior falls within normal parameters but whose COLLECTIVE behavior — when correlated across agents — reveals a coordinated pattern: synchronized timing, complementary resource access, distributed data movement that only adds up to exfiltration when the individual streams are combined.

Defensive implementation: Cross-agent behavioral correlation that operates on the fleet level, not the individual agent level. Aggregate behavioral telemetry across all agents and analyze for three correlation types: temporal correlation (are multiple agents performing actions within the same narrow time window at rates exceeding chance?), resource correlation (are different agents accessing complementary subsets of the same dataset or system, where no single agent touches more than its authorized scope but the union of their accesses covers a target?), and volume correlation (does the sum of individually-normal data transfers across agents add up to an abnormally large aggregate total when summed over a time window?).

Extend from: Network traffic analysis (NTA) tools (Darktrace, Vectra, ExtraHop) already correlate traffic patterns across multiple endpoints. Extending to agent-specific behavioral correlation requires defining agent fleet behavioral features and ingesting agent telemetry into the NTA platform.

Scenario #16 — Systemic Hardening Evasion

Countering principles: #33 (systemwide hardening), #4 (staged escalation), #17 (evasion triggers faster learning)

Detection signature: Anomalous agent behavior that occurs specifically during the transition window between hardened mode and normal mode — after the fever response has been declared resolved but before full operational permissions are restored. The attacker waits for the hardening to relax and then acts in the restoration window.

Defensive implementation: Graduated de-escalation with elevated monitoring during the transition. When systemwide hardening is lifted, permissions are restored in stages (not all at once), and monitoring sensitivity remains elevated for a probationary period after each stage. Full de-escalation requires confirmation that no anomalous behavior occurred during any transition stage. The transition window itself is a high-alert period, not a return to baseline.

Extend from: Incident response runbooks already define post-incident monitoring periods. Formalizing the de-escalation stages as automated SOAR playbook steps with gated transitions adds structure to an existing practice.

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11. Mapping Coverage Matrix

#	Mechanism	Kingdom	Category	Status
1-2	MHC-I + NK Missing Self	Mammal	Detection	[DONE] Done (corrected)
3	Antigen Presentation	Mammal	Detection	[DONE] Done
4	PRRs / Two-Speed Detection	Mammal	Detection	[DONE] Done
5	Immune Memory	Mammal	Memory	[DONE] Done
6	MHC Downregulation (evasion)	Virus/Mammal	Evasion	[DONE] Done

#	Mechanism	Kingdom	Category	Status
7	Molecular Mimicry (evasion)	Virus/Mammal	Evasion	[DONE] Done
8	Complement Cascade	Mammal	Response	[DONE] Done
8e	Virokines (evasion)	Virus/Mammal	Evasion	[DONE] Done
9	Thymic Education	Mammal	Calibration	[DONE] Done
11a	Treg Co-option	Mammal	Regulation	[DONE] Done
11b	Peripheral Tolerance Induction	Mammal	Regulation	[DONE] Done
12	Dendritic Cells	Mammal	Communication	[DONE] Done
13	Inflammation	Mammal	Response	[DONE] Done
14	Immune Privilege	Mammal	Architecture	[DONE] Done
15	Gut Microbiome Tolerance	Mammal/Bacteria	Architecture	[DONE] Done
16	Viral Latency (evasion)	Virus	Evasion	[DONE] Done
17	Cell-Autonomous Security	Plant	Architecture	[DONE] Done
18	Hypersensitive Response	Plant	Response	[DONE] Done
19	Xenobiotic Detoxification (evasion)	Insect/Plant	Evasion	[DONE] Done
20	CRISPR-Cas	Bacteria	Memory	[DONE] Done
21	Transgenerational Immune Priming	Insect/Plant	Memory	[DONE] Done
22	Apoptosis vs. Necroptosis	Mammal	Response	[DONE] Done
23	Interferon Response	Mammal	Communication	[DONE] Done
24	Affinity Maturation	Mammal	Learning	[DONE] Done
25	Trained Immunity	Mammal	Learning	[DONE] Done
26	Antigenic Variation	Pathogen	Evasion	[DONE] Done
27	Intracellular Hiding	Pathogen	Evasion	[DONE] Done
28	Clonal Expansion	Mammal	Response	[DONE] Done
29	Immune Checkpoints (PD-1/CTLA-4)	Mammal	Evasion	[DONE] Done
30	Complement Evasion	Pathogen	Evasion	[DONE] Done
31	Melanization/Encapsulation	Insect	Response	[DONE] Done

#	Mechanism	Kingdom	Category	Status
32	RNA Interference	Plant/Invertebrate	Detection	[DONE] Done
33	Biofilm Formation	Bacteria	Evasion	[DONE] Done
34	Fever	Mammal/Vertebrate	Response	[DONE] Done

Remaining candidates (partially covered by existing mappings):

#	Mechanism	Kingdom	Notes
—	V(D)J Recombination	Mammal	Diversity generation mechanism — the OUTPUT is covered by clonal selection (#28) and affinity maturation (#24); the modular assembly PROCESS could be expanded
—	Allorecognition	Colonial organisms	Colony-level self/non-self — structurally similar to MHC-I (#1-2) at a different organizational scale
—	Cross-Presentation	Mammal	Evidence sharing between detection paths — partially covered by dendritic cells (#12) and antigen presentation (#3)

12. Applications and Future Work

12.1 Core Research Question

The central question this work addresses: *How should enterprise security infrastructure detect and respond to compromised or misbehaving AI agents that hold real credentials and make autonomous decisions inside identity systems?*

This is not a theoretical question. Agentic AI workloads are entering enterprise environments now — holding OAuth tokens, calling APIs, accessing data stores, making tool-use decisions. The security controls designed for human users and static service accounts are structurally inadequate for entities that are autonomous, adaptive, and capable of complex multi-step behavior. The immune system — nature's answer to the same structural problem (detecting compromised autonomous entities operating inside a trusted infrastructure) — provides a rich source of design patterns for this new problem space.

12.2 What This Work Contributes

A methodology. Biomimetic gap analysis — cataloging nature's solutions as abstract structural patterns, overlaying them against a technology problem space, and identifying gaps — is a reusable approach that extends beyond agent security. The same methodology could be applied to any domain where adversarial detection is the core problem: fraud detection, content moderation, network intrusion detection, supply chain integrity.

A detection philosophy. The "motion detector" concept — watching what agents DO rather than checking their credentials — emerges consistently across multiple immune system mappings. MHC-I behavioral sampling, NK cell missing-self detection, and the corrected three-state model all converge on the same principle: behavioral evidence, infrastructure-sampled, is more reliable than identity assertions for detecting compromise in autonomous entities.

A catalog of attack scenarios. The 16 evasion-derived attack scenarios are grounded in pathogen strategies that have been refined by billions of years of adversarial evolution. Several — including the sleeper agent (viral latency), baseline drift exploitation (peripheral tolerance induction), defense neutralization (xenobiotic detoxification), digital biofilm (coordinated collective evasion), checkpoint exhaustion (alert fatigue weaponization), and trusted boundary exploitation (intracellular hiding) — do not appear in existing agent security literature in this form. These scenarios can inform red-team exercises, threat modeling, and security architecture review for organizations deploying agentic AI.

An architectural framework. The 33 design principles, drawn from detection systems across six kingdoms and domains of life (mammals, plants, bacteria, insects,

pathogenic protists, and viruses), constitute a coherent philosophy for building agent monitoring infrastructure. Key architectural insights — cell-autonomous security from plants, heritable threat memory from CRISPR, continuous tolerance maintenance from the gut microbiome, detection-as-weapon from RNA interference, active hostile containment from insect melanization, and systemwide environmental hardening from the fever response — offer genuinely different paradigms from the centralized monitoring models that dominate current security architecture.

12.3 The Practical Artifact

The coffer-mcp project (github.com/annawhooo/coffer-mcp) — an MCP server for encrypted credential management with AES-256-GCM encryption, OS keyring integration, URL allowlisting, and audit logging — is the concrete artifact that first exposed the gap this research addresses. Building a tool that gives AI agents access to real credentials made the detection problem tangible: if an agent holds a credential and misbehaves, what detects that? The immune system mappings provide the theoretical framework; coffer-mcp provides the practical ground truth that motivated the question.

12.4 Future Work

Prototype implementation. Select 2-3 design principles and implement them as proof-of-concept monitoring capabilities against agentic workloads. Priority candidates: infrastructure-sampled behavioral signatures (Mapping #3), heritable threat memory (Mapping #20), cell-autonomous self-assessment (Mapping #17), and detection-as-response (Mapping #32, RNAi).

Validation against real attack patterns. Test the 16 biologically-derived attack scenarios against actual agent frameworks (LangChain, AutoGPT, MCP-based agents) to determine which are practically exploitable today and which require capabilities not yet available to attackers.

Immune failure mode analysis. This document draws on immune system successes. But biological immune systems fail constantly — autoimmune disease (attacking self → false positive damage), cancer immune evasion (missing a threat → APT persistence), chronic infection (failing to clear → persistent compromise), and cytokine storm (over-response → automated systems causing more damage than the original threat). Systematic analysis of immune failure modes would yield additional defensive insights and help identify where the design principles could themselves cause harm if miscalibrated.

Cross-domain methodology generalization. Apply the biomimetic gap analysis methodology to a second problem domain (e.g., financial fraud detection, content moderation) to validate that the approach is domain-independent.

Formal peer review. Submit for review by both immunologists and security researchers to identify errors in cross-domain translation. The biological descriptions in this document simplify complex mechanisms to extract structural patterns; immunologist review would ensure no simplification introduces a materially incorrect mapping.

Expand remaining candidate mechanisms. Three mechanisms are partially covered by existing mappings but could be expanded: V(D)J recombination (the modular diversity generation PROCESS, distinct from the diversity OUTPUT already covered in clonal selection and affinity maturation), allorecognition in colonial organisms (colony-level self/non-self, relevant to multi-agent fleet trust), and cross-presentation (evidence sharing between detection paths that normally don't share data). Additionally, social insect collective immunity (ant colonies, termite mounds) is an untapped source for multi-agent coordination patterns.

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13. Limitations and Honest Caveats

Immune systems fail. This document draws on immune system successes — effective detection mechanisms, elegant architectural solutions. But biological immune systems fail constantly: autoimmune disease (attacking self), cancer immune evasion (missing a threat), chronic infection (failing to clear a pathogen), immunodeficiency (inadequate response), and cytokine storm (over-response that kills the host). The immune system is an IMPERFECT solution to the adversarial detection problem. Its failures are as instructive as its successes — autoimmune disease maps directly to alert fatigue and false positive damage; cancer immune evasion maps to APT persistence; cytokine storm maps to automated response systems causing more damage than the original threat. A complete treatment should analyze failure modes with equal rigor.

Analogical reasoning is a heuristic, not a proof. Every mapping in this document is a hypothesis generated by structural analogy. Structural similarity between domains suggests that a solution or vulnerability MAY transfer, but it does not guarantee it. Each mapping requires independent validation against real agentic infrastructure before it can be considered actionable.

The locality constraint is fundamental. Many immune system strategies depend on physical locality — threats must travel, signaling has speed limits, containment is spatial. Digital systems lack these constraints. Every containment-oriented mapping should be evaluated against the question: "Does this pattern depend on physical locality?" If yes, the digital version may need fundamental redesign, not direct translation.

Gap claims are practitioner assessments. The claims about what does or does

not exist in current agent security practice are based on the authors' professional experience in enterprise identity and security architecture. They are not the product of a systematic survey of all existing agent security frameworks. "To the authors' knowledge" should be understood as qualifying all gap claims.

Species-specific accuracy vs. pattern-level accuracy. Some biological descriptions in this document simplify complex mechanisms to extract the structural pattern. The simplified descriptions are adequate for the cross-domain mapping purpose but should not be cited as authoritative immunology. Readers seeking biological accuracy should consult the primary sources listed below.

Responsible disclosure tension. This paper derives 16 specific attack scenarios from biological evasion strategies. Several of these — particularly telemetry suppression (#1), checkpoint exhaustion (#14), digital biofilm (#15), and identity rotation (#12) — describe attack patterns that could be executed against existing agent frameworks. Publishing specific attack mechanisms without corresponding defensive implementations creates asymmetric value: defenders receive design principles (which require engineering investment to implement), while attackers receive operational blueprints (which require less investment to execute). The authors acknowledge this tension. The design principles and risk-weighted prioritization (Section 9.1) are intended to give defenders actionable guidance, and future work will focus on reference implementations that close the highest-priority gaps. Until those implementations exist, the attack scenarios in this paper should be treated as threat modeling inputs, not as penetration testing guides.

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14. Author Contributions

Anna Hix conceived the research framing (behavioral signatures of compromised agents as observable phenomena in identity infrastructure), developed the biomimetic gap analysis methodology, authored 33 of 34 mappings, derived all 33 design principles and 16 attack scenarios, conducted the prior art analysis, and wrote the manuscript. Research context: The gap this paper addresses was first identified during development of coffer-mcp (github.com/annawhooo/coffer-mcp), a credential management tool for MCP servers. Building coffer-mcp revealed that the agent security ecosystem has no mechanism for detecting when an agent holding real credentials begins misbehaving — the "motion detector problem" that MHC-I behavioral sampling (Mapping #1-2) directly addresses. coffer-mcp is the symptom that exposed the gap, not itself a solution to it.

Shaun Milligan contributed Mapping #19 (Xenobiotic Detoxification → Active Defense Neutralization), derived from professional expertise in environmental management and landscaping — specifically, the observation that caterpillars actively neu-

tralize plant defensive chemicals rather than merely tolerating them. This cross-domain insight exemplifies the value of non-specialist perspectives in structural pattern identification. Milligan also identified the determinate/indeterminate analogy (Section 2.0.1) that provides the methodological justification for biological immunity as the source domain: the observation that the distinction between determinate and indeterminate plant species structurally parallels the distinction between deterministic and non-deterministic AI systems, and that biological immune systems evolved specifically to monitor indeterminate (non-deterministic) systems — the exact problem that current security tooling fails to address for agentic AI.

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16. References

[1] Janeway CA Jr, Travers P, Walport M, et al. *Immunobiology: The Immune System in Health and Disease.* 5th ed. New York: Garland Science; 2001.

[2] "MHC class I." Wikipedia. Accessed April 2026. https://en.wikipedia.org/wiki/MHC_class_I

[3] "Innate Immune Response." In: *Concepts of Biology — 1st Canadian Edition.* OpenStax/BCcampus; 2015. https://opentextbc.ca/biology/chapter/23-1-innate-immune-response/

[4] Raulet DH. "Missing self recognition and self tolerance of natural killer (NK) cells." *Semin Immunol.* 2006;18(3):145-150.

[5] Kim S, Poursine-Laurent J, Truscott SM, et al. "Licensing of natural killer cells by host major histocompatibility complex class I molecules." *Nature.* 2005;436:709-713.

[6] Bryceson YT, Long EO. "A new self: MHC-class-I-independent natural-killer-cell self-tolerance." *Nat Rev Immunol.* 2005;5:363-374.

[7] Held W. "Adaptations of natural killer cells to self-MHC class I." *Front Immunol.* 2014;5:349.

[8] "MHC class I and MHC class II molecules." Osmosis. https://www.osmosis.org/learn/MHC_class

[9] Warrington R, Watson W, Kim HL, Antonetti FR. "An introduction to immunology and immunopathology." *Allergy Asthma Clin Immunol.* 2011;7(Suppl 1):S1.

[10] Wang Q, Peng C, Yang M, et al. "MHC-I pathway disruption by viruses: insights into immune evasion and vaccine design for animals." *Front Immunol.* 2025;16:1540159.

[11] Alcami A, Koszinowski UH. "Viral mechanisms of immune evasion." *Trends Microbiol.* 2000;8(9):410-418.

[12] Zamora-Ceballos M, et al. "Molecular mimicry as a mechanism of viral immune evasion and autoimmunity." *Nat Commun.* 2024;15:9403.

[13] Schmid-Hempel P. "Immune defence, parasite evasion strategies and their relevance for 'macroscopic phenomena' such as virulence." *Philos Trans R Soc Lond B Biol Sci.* 2009;364(1513):85-98.

[14] "The complement system and innate immunity." In: Janeway CA Jr, et al. *Immunobiology.* 5th ed. Garland Science; 2001. NCBI Bookshelf NBK27100.

[15] Merle NS, Church SE, Fremeaux-Bacchi V, Roumenina LT. "Complement system part II: role in immunity." *Front Immunol.* 2015;6:257.

[16] "Complement system." Wikipedia. Accessed April 2026. https://en.wikipedia.org/wiki/Comple

[17] Bardhan M, Kaushik R. "Physiology, Complement Cascade." In: *StatPearls.* Treasure Island (FL): StatPearls Publishing; 2023.

[18] "T cell." Wikipedia. Accessed April 2026. https://en.wikipedia.org/wiki/T_cell

[19] Takahama Y, Ohigashi I, Baik S, Anderson G. "Thymus machinery for T-cell selection." *Int Immunol.* 2019;31(3):119-125.

[20] Burroughs NJ, et al. "Self-mediated positive selection of T cells sets an obstacle to the recognition of nonself." *Proc Natl Acad Sci USA.* 2021;118(38):e2100542118.

[21] Klein L, Kyewski B, Allen PM, Hogquist KA. "Positive and negative selection of the T cell repertoire: what thymocytes see and don't see." *Nat Rev Immunol.* 2014;14(6):377-391.

[22] Kurd NS, Lutes LK, Yun TJ, et al. "T cell self-reactivity during thymic development dictates the timing of positive selection." *eLife.* 2021;10:e65435.

[23] Takaba H, Takayanagi H. "The mechanisms of T cell selection in the thymus." *Trends Immunol.* 2017;38(11):805-816.

[24] "Innate immune system." In: *Autoimmunity.* NCBI Bookshelf NBK459455; 2013.

[25] "Immune privilege." Wikipedia. Accessed April 2026. https://en.wikipedia.org/wiki/Immune_p

[26] "Immune Privilege." ScienceDirect Topics. Accessed April 2026.

[27] London A, Benhar I, Schwartz M. "The retina as a window to the brain — from eye research to CNS disorders." *Nat Rev Neurol.* 2013;9(1):44-53. See also: Forrester JV, Xu H. "The privileged immunity of immune privileged organs: the case of the eye." *Front Immunol.* 2012;3:296.

[28] "Puzzling over privilege: How the immune system protects—and sometimes doesn't." McGill University, Dept of Medicine; 2018.

[29] Zhao S, Zhu W, Xue S, Han D. "Testicular defense systems: immune privilege and innate immunity." *Cell Mol Immunol.* 2014;11(5):428-437. See also: Li N, Wang T, Han D. "Structural, cellular and molecular aspects of immune privilege in the testis." *Front Immunol.* 2012;3:152.

[30] Wiertsema SP, van Bergenhenegouwen J, Garssen J, Knippels LMJ. "The interplay between the gut microbiome and the immune system in the context of infectious diseases throughout life and the role of nutrition in optimizing treatment strategies." *Nutrients.* 2021;13(3):886.

[31] Belkaid Y, Hand TW. "Role of the microbiota in immunity and inflammation." *Cell.* 2014;157(1):121-141.

[32] Honda K, Littman DR. "The microbiome in infectious disease and inflammation." *Annu Rev Immunol.* 2012;30:759-795. See also: Hooper LV, Littman DR, Macpherson AJ. "Interactions between the microbiota and the immune system." *Science.* 2012;336(6086):1268-1273.

[33] Kamada N, Chen GY, Inohara N, Núñez G. "Control of pathogens and pathobionts by the gut microbiota." *Nat Immunol.* 2013;14(7):685-690.

[34] "Microbial symbiosis and immunity." Wikipedia. Accessed April 2026. https://en.wikipedia.org/wiki/Microbial_symbiosis_and_immunity

[35] Shi N, Li N, Duan X, Niu H. "Interaction between the gut microbiome and mucosal immune system." *Mil Med Res.* 2017;4:14.

[36] Grinde B. "Herpesviruses: latency and reactivation — viral strategies and host response." *J Oral Microbiol.* 2013;5:22766.

[37] Cohen JI. "Herpesvirus latency." *J Clin Invest.* 2020;130(7):3361-3369.

[38] Levy D. "Viruses have the ability to enter a latent phase within their host." *J Microb Pathog.* 2023;7:160.

[39] Goldberg GW, et al. “Bacteria exploit viral dormancy to establish CRISPR-Cas immunity.” *Cell Host Microbe.* 2025;33(3).

[40] Shine MB, Xiao X, Kachroo P, Kachroo A. “The lifecycle of the plant immune system.” *Mol Plant.* 2019;12(12):1541-1570.

[41] Jones JDG, Dangl JL. “The plant immune system.” *Nature.* 2006;444:323-329.

[42] Vlot AC, Sales JH, Lenk M, et al. “Systemic propagation of immunity in plants.” *New Phytol.* 2021;229:1234-1250.

[43] “Systemic acquired resistance.” Wikipedia. Accessed April 2026. https://en.wikipedia.org/wik

[44] Ali S, Ganai BA, Kamili AN, et al. “Modulation of plant defense system in response to microbial interactions.” *Front Microbiol.* 2020;11:1298.

[45] Zhou Y, et al. “Transcriptomic and metabolomic profiling reveal the mechanism of tomatine resistance in *Spodoptera litura.*” *Front Physiol.* 2019;10:8.

[46] Peng T, et al. “Functional study of cytochrome P450 enzymes from the brown planthopper to analyze its adaptation to BPH-resistant rice.” *Front Physiol.* 2017;8:972.

[47] Barrangou R, Marraffini LA. “CRISPR-Cas systems: prokaryotes upgrade to adaptive immunity.” *Mol Cell.* 2014;54(2):234-244.

[48] Marraffini LA. “CRISPR-Cas immunity in prokaryotes.” *Nature.* 2015;526:55-61.

[49] Rath D, Amlinger L, Rath A, Lundgren M. “The CRISPR-Cas immune system: biology, mechanisms and applications.” *Biochimie.* 2015;117:119-128.

[50] McGinn J, Marraffini LA. “CRISPR-Cas systems optimize their immune response by specifying the site of spacer integration.” *Mol Cell.* 2016;64(3):616-623. See also: Hille F, et al. “The biology of CRISPR-Cas: backward and forward.” *Cell.* 2018;172(6):1239-1259.

[51] Goldberg GW, et al. “Bacteria exploit viral dormancy to establish CRISPR-Cas immunity.” *Cell Host Microbe.* 2025;33(3).

[52] Faure G, Makarova KS, Koonin EV. “CRISPR–Cas: complex functional networks and multiple roles beyond adaptive immunity.” *J Mol Biol.* 2019;431(1):13-33.

[53] Vilcinskas A. “Mechanisms of transgenerational immune priming in insects.” *Dev Comp Immunol.* 2021;124:104205.

[54] Tetreau G, et al. “Trans-generational immune priming in invertebrates: current knowledge and future prospects.” *Front Immunol.* 2019;10:1938.

[55] Mukherjee K, Vilcinskas A. “Epigenetic remodeling in insect immune memory.” *Front Immunol.* 2024;15:1397521.

[56] Gegner J, Baudach A, Muber K, et al. "Epigenetic mechanisms are involved in sex-specific trans-generational immune priming in the lepidopteran model host *Manduca sexta*." *Front Physiol.* 2019;10:137.

[57] Sadd BM, Schmid-Hempel P. "Insect immunity shows specificity in protection upon secondary pathogen exposure." *Curr Biol.* 2006;16(12):1206-1210. See also: Sadd BM, et al. "Trans-generational immune priming in a social insect." *Biol Lett.* 2005;1(4):386-388.

[58] Trauer-Kizilelma U, Hilker M. "Transcriptomics reveals specific molecular mechanisms underlying transgenerational immunity in *Manduca sexta*." *Ecol Evol.* 2020;10:11251-11261.

[59] Sheridan C. "How plants and insects inherit immunity from their parents." *Nature.* 2019;575:S58-S59.

[60] AskNature / Biomimicry Institute. "Biomimicry Taxonomy." https://asknature.org/resource/bio
taxonomy/

[61] Deldin JM, Schuknecht M. "The AskNature Database: Enabling Solutions in Biomimetic Design." In: Goel A, McAdams D, Stone R, eds. *Biologically Inspired Design.* London: Springer; 2014:17-27.

[62] Helfman Cohen Y, Reich Y. *Biomimetic Design Method for Innovation and Sustainability.* Springer; 2016. See also: FindStructure database, http://www.findstructure.org/

[63] Hofstadter DR, Sander E. *Surfaces and Essences: Analogy as the Fuel and Fire of Thinking.* New York: Basic Books; 2013.

[64] Gentner D. "Analogy." *Open Encyclopedia of Cognitive Science.* MIT; 2025.

[65] Truyers C, et al. "Overlooked sources of inspiration in biomimetic research." *Sci Rep.* 2025;15:11703.

[66] Xu H, et al. "Targeting MHC-I molecules for cancer: function, mechanism, and therapeutic prospects." *Mol Cancer.* 2023;22:194.

[67] Gu W, et al. "Reverse signaling by MHC-I molecules in immune and non-immune cell types." *Front Immunol.* 2020;11:605958.

[68] "Major histocompatibility complex." Wikipedia. Accessed April 2026. https://en.wikipedia.org/wiki/Major_histocompatibility_complex

[69] Sullivan BA, et al. "MHC class Ib molecules bridge innate and acquired immunity." *Nat Rev Immunol.* 2006;6:459-467.

[70] Innate and adaptive immunity: specificities and signaling hierarchies revisited. *Nat Immunol.* 2005. PMC7097365.

Apoptosis and Cell Death

[171] Bertheloot D, Latz E, Franklin BS. "Necroptosis, pyroptosis and apoptosis: an intricate game of cell death." *Cell Mol Immunol.* 2021;18:1106-1121.

[172] Fink SL, Cookson BT. "Apoptosis, pyroptosis, and necrosis: mechanistic description of dead and dying eukaryotic cells." *Infect Immun.* 2005;73(4):1907-1916.

[174] "Necrosis vs. Apoptosis: Key Differences Explained." Proteintech. Accessed April 2026.

[176] Han J, Zhong CQ, Zhang DW. "Programmed necrosis: backup to and competitor with apoptosis in the immune system." *Nat Immunol.* 2011;12:1143-1149.

[177] Green DR, Ferguson T, Zitvogel L, Kroemer G. "Immunogenic and tolerogenic cell death." *Nat Rev Immunol.* 2009;9:353-363.

Interferon Response

[181] "Chasing virus replication and infection: PAMP-PRR interaction drives type I interferon production." *Viruses.* 2025;17(4):528.

[182] Schoggins JW, Rice CM. "Interferon-stimulated genes and their antiviral effector functions." *Curr Opin Virol.* 2011;1(6):519-525. See also: Schneider WM, Chevillotte MD, Rice CM. "Interferon-stimulated genes: a complex web of host defenses." *Annu Rev Immunol.* 2014;32:513-545.

[183] Garcia-Sastre A. "Interferons: signaling, antiviral and viral evasion." *Immunol Rev.* 2011;243(1). PMC7112942.

[185] "Interferon and immunity: the role of microRNA in viral evasion strategies." PMC12101089. 2025.

[188] Ivashkiv LB, Donlin LT. "Regulation of type I interferon responses." *Nat Rev Immunol.* 2014;14:36-49.

Affinity Maturation

[191] Victora GD, et al. "Regulated somatic hypermutation enhances antibody affinity maturation." *Nature.* 2025.

[192] "Affinity maturation." Wikipedia. Accessed April 2026. https://en.wikipedia.org/wiki/Affinity_
— *[Replace with primary source for publication: Victora GD, Nussenzweig MC. "Germinal centers." Annu Rev Immunol. 2012;30:429-457.]*

[196] Tan CS, et al. "Somatic hypermutation generates antibody specificities beyond the primary repertoire." *Immunity.* 2025;58(5).

[198] Bannard O, Cyster JG. "Germinal center reaction: antigen affinity and presentation explain it all." *Trends Immunol.* 2017;38(10). PMC4174395.

Trained Immunity

[201] Ziogas A, et al. "Trained immunity: induction of an inflammatory memory in disease." *Cell Res.* 2025.

[202] "Innate immune memory: The evolving role of macrophages in therapy." *eLife.* 2025;108276.

[204] Netea MG, et al. "Defining trained immunity and its role in health and disease." *Nat Rev Immunol.* 2020;20:375-388.

[205] "Innate immune memory in macrophage differentiation and cardiovascular diseases." PMC12131520. 2025.

[207] Wang Y, et al. "BCG-induced trained immunity: history, mechanisms and potential applications." *J Transl Med.* 2023;21:106.

Antigenic Variation

[211] Horn D. "Antigenic variation in African trypanosomes: the importance of chromosomal and nuclear context in VSG expression control." *Cell Microbiol.* 2014;16(5). PMC3963442.

[212] Morrison LJ, Marcello L, McCulloch R. "Antigenic variation in the African trypanosome: molecular mechanisms and phenotypic complexity." *Cell Microbiol.* 2009;11(12):1724-1734.

[213] Hall JPJ, et al. "Faster growth with shorter antigens can explain a VSG hierarchy during African trypanosome infections: a feint attack by parasites." *Sci Rep.* 2018;8:10922.

[214] Mugnier MR, Stebbins CE, Papavasiliou FN. "Variant surface glycoprotein density defines an immune evasion threshold for African trypanosomes undergoing antigenic variation." *Nat Commun.* 2017;8:828.

[215] Cross GAM, Kim HS, Wickstead B. "Capturing the variant surface glycoprotein repertoire (the VSGnome) of *Trypanosoma brucei* Lister 427." *Mol Biochem Parasitol.* 2014;195(1):59-73.

[216] Zhang X, et al. "A coordinated transcriptional switching network mediates antigenic variation of human malaria parasites." *eLife.* 2022;11:e83840.

Intracellular Hiding

[221] Liu CH, Liu H, Ge B. "Innate immunity in tuberculosis: host defense vs pathogen evasion." *Cell Mol Immunol.* 2017;14(12):963-975. See also: "The immune escape mechanisms of *Mycobacterium tuberculosis.*" PMC6359177. 2019.

[223] Divangahi M, Bhatt K. "Evasion of innate immunity by *Mycobacterium tuberculosis*: is death an exit strategy?" *Nat Rev Microbiol.* 2009;7:773-782. PMC3221965.

[225] "Intracellular survival of microorganisms in macrophages." Biology Insights. 2025.

[226] Chandra P, Grigsby SJ, Philips JA. "Immune evasion and provocation by *Mycobacterium tuberculosis.*" *Nat Rev Microbiol.* 2022;20:750-766.

Clonal Expansion

[231] Kaiser GE. "Clonal selection and clonal expansion." In: *Microbiology.* Biology LibreTexts; 2023.

[233] "Clonal selection." Wikipedia. Accessed April 2026. https://en.wikipedia.org/wiki/Clonal_sele
— *[Replace with primary source for publication: Burnet FM. The Clonal Selection Theory of Acquired Immunity. Cambridge University Press; 1959.]*

[234] Cichocki F, Grzywacz B, Miller JS. "Clonal expansion of innate and adaptive lymphocytes." *Nat Rev Immunol.* 2020;20:694-707.

[236] "Clonal selection and T-cell differentiation." Medicine LibreTexts; 2025.

Immune Checkpoints

[241] Buchbinder EI, Desai A. "CTLA-4 and PD-1 pathways: similarities, differences, and implications of their inhibition." *Am J Clin Oncol.* 2016;39(1):98-106. PMC4892769.

[242] Bruner JT, et al. "Mechanisms and clinical implications of immune checkpoint inhibitors PD-1, CTLA-4, and TIM-3 in cancer." *J Cancer Immunol.* 2025;7(1):20-29.

[243] "Combination of PD-1/PD-L1 and CTLA-4 inhibitors in the treatment of cancer — a brief update." *Front Immunol.* 2025;16:1680838.

[244] "Overcoming resistance to PD-1 and CTLA-4 blockade mechanisms and therapeutic strategies." *Front Immunol.* 2025;16:1688699.

Complement Evasion

[251] Parente R, Clark SJ, Gros P. "Hijacking Factor H for complement immune evasion." *Front Immunol.* 2021;12:602277.

[252] Agrawal P, et al. "Complement evasion strategies of viruses: an overview." *Front Microbiol.* 2017;8:1117.

[253] Lambris JD, Ricklin D, Geisbrecht BV. "Complement evasion by human pathogens." *Nat Rev Microbiol.* 2008;6:132-142.

[254] Rooijakkers SHM, van Strijp JAG. "Molecular mechanisms of complement evasion: learning from staphylococci and meningococci." *Nat Rev Microbiol.* 2007;5:9-21.

[255] "Microbial evasion of the complement system: a continuous and evolving story." *Front Immunol.* 2023;14:1281096.

Melanization/Encapsulation

[261] "How to eliminate pathogen without killing oneself? Immunometabolism of encapsulation and melanization in *Drosophila*." *Front Immunol.* 2023;14:1330312.

[263] "Innate immunity in insects: the lights and shadows of phenoloxidase system activation." *Int J Mol Sci.* 2025;26(3):1320.

[265] "Regulation and function of the melanization reaction in *Drosophila*." *Fly.* 2009;3(1):80-86.

RNA Interference

[271] Agrawal N, Bhatt PVS, Malhotra B, Bhatt D. "RNA interference: biology, mechanism, and applications." *Microbiol Mol Biol Rev.* 2003;67(4):657-685. PMC309050.

[272] Gammon DB, Mello CC. "RNA interference-mediated antiviral defense in insects." *Curr Opin Insect Sci.* 2015;8:111-120. PMC4448697.

[273] Asad S, Mukhtar Z. "RNA interference: promising approach to combat plant viruses." *Int J Mol Sci.* 2022;23(10):5312. PMC9142109.

[274] Bronkhorst AW, van Rij RP. "Antiviral RNAi in insects and mammals: parallels and differences." *Viruses.* 2019;11(5):448. PMC6563508.

[276] Li F, Ding SW. "Virus counterdefense: diverse strategies for evading the RNA-silencing immunity." *Annu Rev Microbiol.* 2006;60:503-531. PMC2693410.

Biofilm Formation

[281] Preda VG, Săndulescu O. "Communication is the key: biofilms, quorum sensing, formation and prevention." *Discoveries.* 2019;7(3):e100. PMC7086079.

[282] Li Y-H, Tian X. "Quorum sensing and bacterial social interactions in biofilms." *Sensors.* 2012;12(3):2519-2538. PMC3376616.

[283] "Biofilm, resistance, and quorum sensing: the triple threat in bacterial pathogenesis." *Microbe.* 2025.

[284] "Targeting the holy triangle of quorum sensing, biofilm formation, and antibiotic resistance in pathogenic bacteria." *Microorganisms.* 2022;10(6):1239. PMC9228545.

Fever

[291] Evans SS, Repasky EA, Fisher DT. "Fever and the thermal regulation of immunity: the immune system feels the heat." *Nat Rev Immunol.* 2015;15(6):335-349. PMC4786079.

[292] Boltaña S, et al. "Fever integrates antimicrobial defences, inflammation control, and tissue repair in a cold-blooded vertebrate." *eLife.* 2023;12:e83644. PMC10014077.

[293] Walter EJ, Hanna-Jumma S, Carraretto M, Forni L. "Physiology, Fever." In: *StatPearls.* Treasure Island, FL: StatPearls Publishing; 2023. NBK562334.

[294] Wu J, et al. "Cold-blooded vertebrate utilizes behavioral fever to alleviate T cell apoptosis and optimize antimicrobial immunity." *Proc Natl Acad Sci USA.* 2024;121(51):e2408969121.

Related Work / Prior Art

[301] Farmer JD, Packard NH, Perelson AS. "The immune system, adaptation, and machine learning." *Physica D.* 1986;22(1-3):187-204.

[302] Forrest S, Perelson AS, Allen L, Cherukuri R. "Self-nonself discrimination in a computer." In: *Proceedings of the 1994 IEEE Symposium on Research in Security and Privacy.* IEEE; 1994:202-212.

[303] de Castro LN, Von Zuben FJ. "Learning and optimization using the clonal selection principle." *IEEE Trans Evol Comput.* 2002;6(3):239-251.

[304] Greensmith J, Aickelin U, Cayzer S. "Introducing dendritic cells as a novel immune-inspired algorithm for anomaly detection." In: *Artificial Immune Systems (ICARIS).* Springer; 2005:153-167.

[305] de Castro LN, Timmis J. *Artificial Immune Systems: A New Computational Intelligence Approach.* London: Springer; 2002.

[306] Sabitha R, Gopikrishnan S, Bejoy BJ, et al. "Network based detection of IoT attack using AIS-IDS model." *Wirel Pers Commun.* 2023;128(3):1543-1566.

[307] Hosseini E. "Artificial immune systems for industrial intrusion detection: a systematic review and conceptual framework." *J Eng.* 2025;2025:8408209.

[308] Anderson W, Moore K, Ables J, et al. "Leveraging artificial immune systems for semi-supervised intrusion detection." In: *CSCI 2024*. Communications in Computer and Information Science, vol 2510. Springer; 2025.

[309] Schrom E, Kinzig A, Forrest S, Graham AL, Levin SA, Bergstrom CT, Castillo-Chavez C, Collins JP, de Boer RJ, Doupé A, Ensafi R, Feldman S, Grenfell BT, Halderman JA, Huijben S, Maley C, Moses M, Perelson AS, Perrings C, Plotkin J, Rexford J, Tiwari M. "Challenges in cybersecurity: Lessons from biological defense systems." *Math Biosci.* 2023;362:109024.

[310] Neelou E, Novikov I, Moroz M, Narayan O, Saade T, Ayenson M, Kabanov I, Ozmen J, Lee E, Narajala VS, Guilherme E, Huang K, Gulsin H, Ross J, Vyshegorodtsev M, Travers A, Habler I, Jadav R. "A2AS: Agentic AI Runtime Security and Self-Defense." SSRN 5579570. September 2025.

[311] Evani PK. "Agentic AI Security: A Control Framework for Autonomous Decision-Making Systems." SSRN 5332681. July 2025.

[312] Tutuncuoglu BT. "SentinelMind: A Self-Learning Autonomous Framework for Real-Time Cyber Threat Anticipation and Response." SSRN 5377248. June 2025.

[313] Tallam K. "The Cyber Immune System: Harnessing Adversarial Forces for Security Resilience." arXiv:2502.17698. February 2025.

[314] "Open Challenges in Multi-Agent Security: Towards Secure Systems of Interacting AI Agents." arXiv:2505.02077. May 2025.